

NEURAL BELLMAN OPERATORS

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We study neural policy evaluation, improvement, and independent verification in a continuous economy with costly preference adjustment. An affine-preference market-deflator dual and validated policy evaluation retain the original utility, adaptive controls, and first-exit contract. A constructive cost-continuation theorem turns eighteen certified policies into a policy rule with regret below .01 for every cost coefficient in $[\cdot 5, 8]$ at the central initial state. Convex value chords, sign-aware expenditure bounds, and exact envelope breakpoints provide a finite acceptance test for this continuum guarantee. Certified access and cost-comparison intervals identify economic effects separately from policy expenditures. Quantitative output-error allowances, interval neural differentiation, and a structured global action oracle connect implemented policies to verified quantities. Frozen method-specific tuning, complete holdout evidence, and reference-solvable control problems retain separate roles: relative performance, robustness, and absolute-accuracy frontiers. Full proofs and preservation of prior manuscripts and unsuccessful calculations accompany the new construction.

KEYWORDS: Stochastic control, Neural policy iteration, Endogenous preferences, Recursive utility, Verification, Computational economics.

1. INTRODUCTION

A numerical decision rule is useful for economic analysis only when its error is small relative to the comparison being made. This requirement is distinct from a small training loss. It is also distinct from outperforming another learning algorithm. When a decision maker can change a preference state, an apparent welfare gain may reflect either the adjustment technology or an inaccurate evaluation of the policy without that technology. When termination is an economic contract, an inaccurate treatment of first exit can dominate both effects.

This paper studies Neural Bellman Operators (NBO): separated policy evaluation and feasible policy improvement, implemented by neural networks and assessed by independent verification. The motivating economy combines consumption, portfolio choice, stochastic risk aversion, and costly preference adjustment. The analysis retains its original running utility, continuous controls, correlated Brownian shocks, and first-exit liquidation covenant. Neither reflecting the state at a training boundary nor altering the running payoff is used to establish the economic results below.

The main numerical result is a certified .01-optimal policy in the original continuous economy at the central initial state, for adjustment-cost coefficients .5, 2, and 8. The respective regret upper bounds are .00987440, .00971727, and .00996896. Each lower endpoint evaluates a feasible policy; each upper endpoint covers all original adaptive controls. The computation does not replace the first-exit contract, restrict the original optimum to time controls, or use a finite action grid to claim a continuous guarantee.

The three original cost certificates can also be turned into a policy rule for an entire cost interval. We construct a library of eighteen certified policies and prove, at the same initial state, a regret bound of .00999403 for every real $k \in [\cdot 5, 8]$. The acceptance test is finite but its guarantee is not restricted to a parameter grid. Convex chords bound the adaptive optimum, and sign-aware expenditure enclosures transfer each feasible policy to all other costs. Exact rational breakpoint tests identify the largest gap between these bounds. This supplies an executable continuation

in an economically meaningful parameter without altering the payoff, control opportunities, or stopping contract.

The economic consequences are quantitative. Raising the adjustment-cost coefficient from two to eight lowers optimal welfare by an amount in $[\text{.02318257}, \text{.04286880}]$. The width of this interval is smaller than its lower endpoint. Access to adjustment at coefficient two raises optimal welfare by an amount in $[\text{.04096859}, \text{.05973218}]$ relative to the *optimal* no-adjustment economy, rather than relative to a selected no-adjustment rule. Discounted expenditures of the deployed policies are also enclosed independently. The optimal expenditure itself is bounded by valid secants, not identified by relabeling a learned policy as an optimizer.

The computational argument separates policy generation from proof. A learned time-control actor is polished within the feasible output class and evaluated by Gaussian polynomial moments under exact first exit. An affine-preference market-deflator witness then bounds the flexible optimum. Portfolio terms cancel for every adapted original portfolio rule. A supporting source inequality retains flexible adjustment, and a conditional-variance correction pays explicitly for its random marginal value. Outward weighted Taylor boxes enclose the remaining one-Gaussian integrals. Explicit continuation potentials cover artificial verification faces without altering the economic domain.

These constructions improve on the preceding wealth-independent witness, whose $k = 2$ regret upper was .10634 . That witness and its complete proofs remain available; the new certificate is not a relabeling of its evaluation precision as optimality precision. The restricted economy's separate market-deflator bound is retained, together with its independently evaluated feasible policy. It gives an initial restricted-value interval narrower than .00905 . Combining the two economies' bounds establishes the welfare inequalities above.

A second contribution addresses the inner optimization needed for verification. In a coupled nonconvex control family, all nonconvex coupling depends on the sum of the actions. A one-dimensional branch-and-bound calculation, together with strongly convex scalar dual subproblems, certifies the full continuous action box without a tensor action grid. Executed certificates reach gaps below 10^{-5} through 128 action dimensions. At the eight- and sixteen-dimensional learned-critic test states, interval differentiation of the actual network weights also encloses the error in the floating-point critic gradient. This is an implemented bridge between learned parameters and an action certificate, not an assumption that an optimizer's stationary point is globally greedy.

The same nonconvex family is given an absolute accuracy axis. A pointwise running-cost relaxation and removal of control bounds admit a Cole–Hopf representation that retains the original nonquadratic terminal payoff. Orthogonal Gaussian coordinates integrate analytically, leaving a positive scalar integral with an outward enclosure. The resulting number is a lower bound for the unchanged original minimization problem. Relative performance is evaluated separately in a frozen holdout experiment: three dimensions, two training budgets, twelve seed pairs per panel, equal learning-rate search budgets, warmed initialization, randomized method order, and a third, independently specified feedback rule. The pinned author implementation of SOC-MartNet is not modified.

Classical verification, monotone approximation, and primal–dual numerical methods supply important foundations. The stopped Itô argument is not claimed as a new general verification principle. Nor is alternating neural evaluation and improvement claimed as a new algorithmic idea. The contribution is the constructive combination of original-contract verification, independent economic upper and lower bounds, computable neural-jet allowances, and a non-enumerative global action audit, with each error quantity matched to an executed calculation. Section 11 compares these objects theorem by theorem with neighboring methods. Recursive

utility, sophisticated temporal selves, dynamic games, stochastic-trace identities, counterexamples, and the complete preceding manuscripts are preserved in the supplement rather than being used to enlarge the empirical claims of the present application.

The present revision also tests two previously separate parts of the numerical argument. A new, prospectively frozen comparison expands method-specific tuning under equal nominal training opportunities, implements a predeclared boundary-extension rule, and uses entirely new holdout seeds. Independently reference-solvable tracking, mean-reverting, and ill-conditioned control problems give absolute-accuracy frontiers for implemented held-action actors. A two-error identity separates deployment discretization from actual actor output error. All original-economy results and the complete earlier external holdout remain visible, so neither a stronger comparison protocol nor an easier reference family can silently replace the motivating economic calculation.

2. THE ECONOMIC ENVIRONMENT

The state is $s = (u, x)$, where u denotes relative risk aversion and x denotes wealth. An admissible action is $a = (c, \theta, p)$ in the compact set

$$A = [.05, .8] \times [-.2, .2] \times [-.5, .8].$$

Controls are progressively measurable and have well-defined stopped state equations. The feedbacks and time controls actually evaluated below are bounded and satisfy this requirement. Before termination,

$$du_t = \theta_t dt + .05 dZ_t^u, \quad (1)$$

$$dx_t = [(.02 + .06p_t)x_t - c_t]dt + .2p_t x_t dZ_t^x, \quad d\langle Z^u, Z^x \rangle_t = -.25 dt. \quad (2)$$

The horizon is $T = 1$, discounting is $\rho = .04$, and the open state domain is $D = (1.2, 2.8) \times (.5, 2)$. Write $\tau = T \wedge \inf\{r \geq t : S_r \notin D\}$. For $k > 0$, running utility and settlement are

$$\ell_k(u, c, \theta) = \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2, \quad (3)$$

$$G(u, x) = -.02(u - 2)^2 + .1 \log x, \quad g(t, u, x) = G(u, x) - 8(1 - t). \quad (4)$$

The policy payoff and optimal value are

$$J_k^\pi(t, s) = \mathbb{E}_{t,s}^\pi \left[\int_t^\tau e^{-\rho(r-t)} \ell_k(S_r, a_r) dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \right], \quad V_k = \sup_\pi J_k^\pi. \quad (5)$$

The no-adjustment value V_R restricts θ to zero and retains *all* admissible consumption and portfolio choices. It is independent of k . The cost experiment holds the controls, transition law, stopping rule, settlement, utility normalization, and consumption units fixed. A state-dependent transformation of utility would generally define another economy.

The early settlement fee is a liquidation covenant. There is no continuation consumption or wealth injection following exit. At the lower wealth boundary the largest admissible wealth drift is negative. At the upper boundary, $p = 0$ eliminates the wealth diffusion and permits an inward drift. Thus neither a silently viable rectangle nor a theorem requiring uniform controlled-boundary regularity can be substituted for this contract. The verification below requires only inequalities for smooth test functions on the trace at which the process is actually stopped.

For a smooth function v , put

$$\mathcal{L}^a v = \theta v_u + [(.02 + .06p)x - c]v_x + .00125v_{uu} + .02p^2x^2v_{xx} - .0025pxv_{ux}, \quad (6)$$

$$R_a v = v_t + \mathcal{L}^a v + \ell_k - \rho v. \quad (7)$$

The mixed derivative is retained. Its sign is not inferred solely from the negative shock correlation.

PROPOSITION 1—Complete selection at a fixed jet: *For the generator in (7), a global maximizing action can be selected componentwise. Consumption is $\text{proj}_{[.05,.8]}(v_x^{-1/u})$ when $v_x > 0$ and is .8 otherwise. Adjustment is $\text{proj}_{[-.2,.2]}(v_u/k)$. Let $h_1 = .06xv_x - .0025xv_{ux}$ and $h_2 = .04x^2v_{xx}$. If $h_2 < 0$, the portfolio is $\text{proj}_{[-.5,.8]}(-h_1/h_2)$; if $h_2 \geq 0$, an endpoint maximizing $h_1p + h_2p^2/2$ is selected, with a fixed tie rule.*

The statement includes convex and zero-curvature portfolio objectives and nonpositive marginal wealth value. It applies to the continuous generator, not to the interpolated, killed-Euler finite-step target. It says nothing about global optimization over shared neural parameters.

3. LEARNING AND VERIFICATION ARE SEPARATE OPERATIONS

For a fixed policy, exact evaluation solves $R_\pi v = 0$ with the stopping contract. Exact improvement maximizes the Hamiltonian of that fixed evaluation. Denote these operators by $\mathcal{E}$ and $\mathcal{I}$; ideal policy iteration is $\pi_{j+1} = \mathcal{I}(\mathcal{E}\pi_j)$. A neural implementation approximates both operators, while an independent evaluator measures the payoff of the policy that will actually be used.

In the raw NBO implementation, critic updates hold the actor fixed. Actor updates hold critic parameters fixed, but retain differentiation through the controlled successor state. A rival's parameters, when present in a game, are also frozen during unilateral improvement. This separation prevents an actor loss from lowering itself by changing the object being evaluated. It does not make the joint network optimization convex. The ideal policy-iteration compactness and limiting arguments are retained in the supplement; no such argument is invoked to certify an observed Adam trajectory.

The inherited finite safeguard is a distinct algorithm. For $\mathcal{T}_n^a w = r_n^a + \delta P_n^a w$, positive sub-stochastic P_n^a , and $\delta = e^{-\rho\Delta}$, define the policy-evaluation residual $e_n = \|v_n - \mathcal{T}_n^{\pi_n} v_{n+1}\|_\infty$ and the full candidate improvement gap $\eta_n = \|\max_a \mathcal{T}_n^a v_{n+1} - \mathcal{T}_n^{\pi_n} v_{n+1}\|_\infty$. If the terminal error is at most b , backward comparison yields

$$\|V_{h,n}^* - V_{h,n}^\pi\|_\infty \leq 2\delta^{N-n}b + \sum_{j=n}^{N-1} \delta^{j-n}(2e_j + \eta_j). \quad (8)$$

The state-dependent envelope and threshold-correction proof are reproduced in the supplement. A threshold can reduce the fraction of decisions changed while leaving the cost of a complete candidate audit essentially unchanged. In particular, the retained 125-action tensor audit is not the non-enumerative certifier of Section 7. Its floating-point envelope is not relabeled as an outward continuous-time bound.

3.1. A quantitative bridge from a fitted network

Let $C_\rho(h) = (1 - e^{-\rho h})/\rho$, with $C_0(h) = h$.

ASSUMPTION 1—Admissible witnesses: *Witnesses are bounded, continuous on the stopped trace, and $C^{1,2}$ in the interior of each time slab. Localized Itô identities pass to the stopping time by uniform integrability. At slab interfaces an upper witness has nonpositive forward jumps and a lower witness has nonnegative forward jumps. The implemented policy has a well-defined stopped state equation.*

THEOREM 1—Stopped one-sided comparison: *Suppose Assumption 1 holds, $U \geq g - b_U$ and $L \leq g + b_L$ on the stopped trace, $\sup_a R_a U \leq \epsilon_U$, and $R_\pi L \geq -\epsilon_L$, where all four allowances are nonnegative. Then*

$$J_k^\pi \geq L - b_L - C_\rho(T - t)\epsilon_L, \quad V_k \leq U + b_U + C_\rho(T - t)\epsilon_U, \quad (9)$$

$$0 \leq V_k - J_k^\pi \leq U - L + b_U + b_L + C_\rho(T - t)(\epsilon_U + \epsilon_L). \quad (10)$$

The inequalities do not require an attained optimizer or a smooth optimal value.

THEOREM 2—Cover-and-oracle certificate for neural output: *Let a continuous neural witness v and a feasible neural policy π satisfy Assumption 1, with stopped-trace error at most b . Let cells cover the full audited time–state domain and have representatives z_j . Suppose independent calculation proves, throughout cell j ,*

$$|R_\pi v(z) - R_\pi v(z_j)| \leq \omega_j, \quad |\Gamma(z) - \Gamma(z_j)| \leq \nu_j, \quad \Gamma = \sup_a R_a v - R_\pi v.$$

Suppose the representative evaluation intervals imply $|R_\pi v(z_j)| \leq e_j$ and a global action oracle implies $\Gamma(z_j) \leq q_j$, including jet, inner-solve, and arithmetic allowances. Set $e = \max_j (e_j + \omega_j)$ and $q = \max_j (q_j + \nu_j)$. Then

$$0 \leq V_k - J_k^\pi \leq 2b + C_\rho(T - t)(2e + q). \quad (11)$$

In particular, certified derivative bounds M_j, N_j and cell radii h_j permit $\omega_j = M_j h_j$ and $\nu_j = N_j h_j$. Any sequence of learned outputs for which these verified allowances vanish has vanishing regret, irrespective of whether its parameters converge.

This is a quantitative a posteriori connection to network errors, not a claim that an empirical mean-square loss controls a supremum. Interval differentiation or weight-based derivative bounds can supply the cell moduli. They may be conservative, and covering a high-dimensional state domain remains costly. Section 7 executes the jet-to-action part for actual learned critics. The economic calculation instead fits an independent polynomial witness; it does not falsely claim to have certified a whole-state neural critic merely because its action audit passed at selected states.

4. A CONSTRUCTIVE CERTIFICATE FOR THE FLEXIBLE ECONOMY

The principal numerical target is a regret bound of .01 for the original problem (5), not for a finite approximation. The certificate below attains this target at $(0, 2, 1.25)$ for each of the three adjustment prices. It uses a feasible policy to bound value from below and a separate, affine-preference market-deflator construction to bound the optimum from above. The latter retains arbitrary adapted consumption, portfolio choice, and preference adjustment. Neither a state lattice nor an optimal-value label enters its construction.

4.1. An affine source that preserves the original opportunities

Set $\phi(u) = -.02(u - 2)^2$, $h = 1 - t$, and let Y satisfy (28), with its initial level $y_0 > 0$ now chosen as a witness parameter. Define

$$S_0(u, y) = \Psi(u, y) + .01y - .004 \log(.5), \quad (12)$$

$$\beta(u, y) = \partial_u \Psi(u, y), \quad (13)$$

$$m(t) = 2 + \int_0^t \bar{\theta}_s ds, \quad 0 \leq \bar{\theta}_s \leq .2, \quad (14)$$

where Ψ is the original consumption conjugate in (30). The reference drift $\bar{\theta}$ is deterministic. It is a parameter of an upper witness, not a restriction on the decision maker's adjustment policy. Write $\hat{y} = \text{proj}_{[.2, 12]} y$ and

$$Q(u) = \phi(m(1)) + \phi'(m(1))(u - m(1)), \quad b_T = \phi'(m(1)).$$

The verification rectangle is $u \in [1.5, 2.5]$, $y \in [.2, 12]$. On it, the following supporting inequality holds:

$$S_0(u, y) \leq S_0(m, y) + \beta(m, y)(u - m), \quad 2 \leq m \leq 2.2. \quad (15)$$

It is a one-sided inequality on the whole rectangle, not a fitted tangent at sampled states. Its proof combines derivative identities and interval inequalities with explicit margins; see Appendix C.1. Global concavity of S_0 outside this rectangle is not assumed.

Put $\hat{S}_r(u, y) = S_0(m(r), \hat{y}) + \beta(m(r), \hat{y})(u - m(r))$. Consider the full-line auxiliary control problem

$$A_k(t, u, y) = \sup_{|\theta| \leq .2} \mathbb{E}_{t, u, y} \left[\int_t^1 e^{-\rho(r-t)} \{ \hat{S}_r(U_r, Y_r) - k\theta_r^2/2 \} dr + e^{-\rho(1-t)} Q(U_1) \right]. \quad (16)$$

Here $dU_r = \theta_r dr + .05 dZ_r^u$, and the auxiliary control remains adapted. The affine source makes its marginal value of u an explicitly defined conditional expectation, which can be bounded without solving another neural control problem. Put

$$Z_u(h, u) = 8h \{ e^{-120(u-1.5)+42h} + e^{-120(2.5-u)+42h} \}, \quad (17)$$

$$Z_y(h, y) = 8h \{ e^{-22 \log(y/.2)+22.33h} + e^{-22 \log(12/y)+22.33h} \}. \quad (18)$$

THEOREM 3—Affine-preference dual and a computable initial upper bound: *For $k \geq .5$, $y_0 \in [1.2, 2.4]$, and any deterministic $\bar{\theta} \in [0, .2]$, the original flexible optimum satisfies*

$$V_k(0, 2, 1.25) \leq .75y_0 + .1 \log(.5) + A_k(0, 2, y_0) + Z_u(1, 2) + Z_y(1, y_0). \quad (19)$$

To evaluate the right side, put $\beta_t = \beta(m(t), \hat{Y}_t)$ and

$$b(s) = \int_s^1 e^{-\rho(t-s)} \mathbb{E} \beta_t dt + e^{-\rho(1-s)} b_T, \quad (20)$$

$$g_k(q, a) = H_k(q) - qa + ka^2/2, \quad (21)$$

$$C = .00375 \int_0^1 t e^{-\rho t} \mathbb{E} \left[\partial_t \beta(m(t), \hat{Y}_t) \right] dt, \quad (22)$$

where the l derivative includes the clipping map and $H_k(q) = \max_{|a| \leq 2} (qa - ka^2/2)$. Then

$$\begin{aligned} A_k(0, 2, y_0) &\leq e^{-\rho} \phi(m(1)) + \int_0^1 e^{-\rho t} \left\{ \mathbb{E} S_0(m(t), \hat{Y}_t) - k \bar{\theta}_t^2 / 2 \right\} dt + C \\ &\quad + \int_0^1 e^{-\rho s} g_k(b(s), \bar{\theta}_s) ds + \frac{.09 D^2}{24k}, \end{aligned} \quad (23)$$

with the explicit bound

$$D^2 = \frac{e^{aM + .045a^2}}{16} \{(M + .09a)^2 + .09\}, \quad a = 12/11, \quad M = \log y_0. \quad (24)$$

All expectations in (22)–(23) are one-Gaussian integrals. They involve no unknown optimal value or policy.

The cancellation of the portfolio is the same accounting identity used in the restricted dual: $d(Yx) = (.04Yx - Yc)dt +$ a local martingale. The new step is to retain flexible adjustment while replacing its source by a supporting affine function. The variance allowance in (23) pays explicitly for the random conditional marginal value of adjustment. Replacing that marginal value by its unconditional mean without this allowance would generally produce an invalid upper bound. The covariance term C also matters: it retains the original correlation between preference and market shocks and is nonpositive here. It is evaluated, not silently discarded or confused with a change of measure.

The source bound $S_0 \geq -7.1$ and $|\beta| \leq 1.1$ imply $A_k - Q \geq -7.925272h$ on the verification rectangle, by choosing zero adjustment in the auxiliary problem. Hence the factors $8h$ in (18) pay for continuation at artificial faces. On actual wealth exits, the uncorrected affine witness already dominates the liquidation settlement. The global fallback $V_k \leq G$ covers continuation after an artificial exit. These facts connect (23) to the original stopped economy rather than to a surrogate full-line economy.

4.2. Policy generation, verified refinement, and the stopping rule

The lower-bound policies start from the retained one-input, twelve-unit neural time-control actors. A constrained L-BFGS-B step polishes their sixteen consumption and adjustment outputs using the original payoff and the same exact budget projection. This is an explicit enlargement from a fixed network's image to the feasible time-control outputs; it is not described as additional Adam iterations or as a theorem that the neural optimizer converges. Consumption and adjustment constants are rounded inward relative to the *exact decimal* action bounds, and feasibility is checked using their exact binary rational values. The finite-node training objective is a proposal mechanism, not an asserted untruncated Gaussian expectation of singular CRRA utility. The independent polynomial-moment evaluator in Appendix I then encloses each deployed continuous-time policy under exact first preference exit. The full network histories, original outputs, polished outputs, and output-optimization records remain separate.

The dual reference is fitted with Gaussian quadrature, but its fitted objective is never used as a bound. Verification uses $t = v^2$, subdivides each of the sixteen time slabs, and covers the standard Gaussian coordinate on $[-6, 6]$. Constant and linear Taylor terms are integrated with

TABLE I
ORIGINAL FLEXIBLE-ECONOMY CERTIFICATES AFTER OUTPUT POLISHING

k	Feasible policy payoff	Optimal-value upper	Regret upper
0.5	$[-1.27267126, -1.27265474]$	-1.26279686	0.00987440
2	$[-1.29502451, -1.29501124]$	-1.28530723	0.00971727
8	$[-1.32817604, -1.32816919]$	-1.31820708	0.00996896

All three regret bounds are below .01. Each upper bound covers the full original continuous control set and stopping contract. Each lower bound evaluates the specified feasible time-control policy. Endpoints are rounded outward; gaps use the unrounded endpoints. These are central-state certificates, not uniform claims over every initial condition.

TABLE II
VERIFIED WORK NEEDED TO CROSS THE ABSOLUTE-ERROR TARGET

k	Source boxes	Regret upper	Verification seconds	.01 met
0.5	4,096	0.01027899	5.46	No
0.5	16,384	0.00995838	11.73	Yes
0.5	65,536	0.00987440	27.54	Yes
2	4,096	0.01013162	5.25	No
2	16,384	0.00980424	11.30	Yes
2	65,536	0.00971727	27.06	Yes
8	4,096	0.01037189	5.29	No
8	16,384	0.01005494	11.48	No
8	65,536	0.00996896	26.79	Yes

The policy is independently evaluated once at the finest stated precision. Time includes that evaluation, dual preparation, and cumulative dual verification through each row, on one CPU thread. It excludes inherited network training, witness fitting, imports, and setup. This is an absolute-accuracy *verification* frontier, not an all-in training comparison or an external matched-accuracy superiority claim. Raw stage clocks are retained.

interval-enclosed weighted moments. Hessian ranges enclose the entire remainder, including boxes crossing the consumption constraint. Gaussian tails, the terminal tangent, the conditional-variance allowance, localization, and outward arithmetic are separately included. The small one-dimensional interval tests establishing (15) precede every certificate execution.

The stopping rule is simple. Given a policy lower bound L and an upper bound U from Theorem 3, return the policy with tolerance ε only when $U - L \leq \varepsilon$. Otherwise refine the quadrature or return the explicit unachieved target at the declared work limit. Soundness does not require optimizer convergence. Nor does soundness imply that an arbitrary tolerance can be reached with a fixed affine relaxation. For the three declared .01 instances, the finite executions below supply the termination certificate.

The finest upper bounds are approximately -1.26279687 , -1.28530724 , and -1.31820708 . Combined with the feasible-policy lower endpoints, they give regret upper bounds .00987440, .00971727, and .00996896 after outward display rounding. The central target is attained without a finite candidate-action grid, an Euler-error assumption, or an assumption that the policy is optimal within its neural class. The preceding wealth-independent witness and its larger gaps are retained in Appendix E, where their loss of wealth information can be compared with the new construction.

TABLE III
SEPARATE UPPER-BOUND ALLOWANCES AT $k = 2$

Boxes	Source remainder	Control gap	Variance allowance	Localization
4,096	0.000339131	0.000063959	0.000081080	0.000000247
16,384	0.000082675	0.000041416	0.000081080	0.000000247
65,536	0.000020398	0.000032100	0.000081080	0.000000247

The source remainder is the Taylor allowance on the covered Gaussian region. Analytic Gaussian tails, interval weight errors, terminal evaluation, and the covariance interval are included in the full records. The variance allowance does not decrease merely by subdividing quadrature boxes. No row is obtained by treating a mesh difference as a rigorous error estimate.

4.3. A quantitative bridge from implemented controls to the certificate

The bound applies to the stored deployment constants themselves. A further perturbation result quantifies its robustness to control evaluation error, rather than inferring robustness from a small training loss.

PROPOSITION 2—Feasible-output error allowance: *Consider two deterministic time-control policies with $p = 0$, $c, \tilde{c} \in [.7, .8]$, $\theta, \tilde{\theta} \in [0, .2]$, and strictly feasible budgets as in (68). Let*

$$\delta_c = \int_0^1 |c_t - \tilde{c}_t| dt, \quad \delta_\theta = \int_0^1 |\theta_t - \tilde{\theta}_t| dt.$$

Their original stopped payoffs satisfy

$$|J_k(c, \theta) - J_k(\tilde{c}, \tilde{\theta})| \leq (3 + .2e^{.02})\delta_c + (25 + .2k + .032)\delta_\theta + 8(14.1 + .02k)e^{-72}. \quad (25)$$

Consequently, a validated output error bounded by the right side is an additive allowance in the returned policy’s regret certificate. Feasibility must still be checked separately; small unconstrained control error alone does not rule out early wealth exit.

For the stored policies the output allowance is zero: their binary64 constants are treated as exact reals by the evaluator. Proposition 2 instead gives an explicit interface for a budget-preserving implementation with perturbed outputs. Together with the neural-jet bound of Theorem 2, it connects observable implementation errors to certified quantities without attributing an unproved global convergence rate to Adam, RMSprop, or L-BFGS-B.

5. CERTIFYING THE VALUE OF PREFERENCE ADJUSTMENT

For each feasible policy write $J_k(\pi) = A(\pi) - kB(\pi)$, where

$$B(\pi) = \mathbb{E}^\pi \int_t^\tau e^{-\rho(r-t)} \theta_r^2 / 2 dr, \quad 0 \leq B(\pi) \leq .02C_\rho(T - t).$$

The policy set is common to every k .

THEOREM 4—Accuracy-calibrated cost comparisons: *The optimal value is convex and nonincreasing in k . If optimizers exist, their adjustment budgets are nonincreasing across distinct prices. More generally, let π_i be ε_i -optimal at k_i , with $k_2 > k_1$. Then*

$$B(\pi_2) - B(\pi_1) \leq \frac{\varepsilon_1 + \varepsilon_2}{k_2 - k_1}. \quad (26)$$

If $V(k) \in [\underline{V}(k), \bar{V}(k)]$, the budget of an optimizer at k , for $k_- < k < k_+$, lies in the primitive budget range intersected with

$$\left[\frac{\underline{V}(k) - \bar{V}(k_+)}{k_+ - k}, \frac{\bar{V}(k_-) - \underline{V}(k)}{k - k_-} \right]. \quad (27)$$

The theorem concerns discounted adjustment expenditures, not a pointwise ordering of controls. A re-solved V_u/k is not a comparative-static proof. The new full-value enclosures in Table I now resolve strict welfare losses across the cost grid. At $k = 2$, (27) also gives the optimal-budget enclosure $[\text{.00386376}, \text{.01960529}]$, conditional on existence of an optimizer. Its positive lower endpoint is informative, while its upper endpoint is still the primitive bound. The tightly enclosed expenditures in Table V describe the specified learned-and-polished policies, not an identified optimal budget.

5.1. A market-deflator witness for the restricted optimum

A tighter comparison becomes possible by exploiting the restricted economy's structure. Introduce the auxiliary process

$$dY_t = .02Y_t dt - .3Y_t dZ_t^x, \quad Y_0 = 1.55. \quad (28)$$

It is a test-process constructed from the original shocks. It does not restrict portfolio information, change wealth, or add a traded asset. For

$$W_0(u, x, y) = y(x - .5) + .1 \log(.5) - .02(u - 2)^2,$$

the p terms in the generator of Yx cancel exactly. With $\theta = 0$, define

$$\Psi(u, y) = \max_{.05 \leq c \leq .8} \left\{ \frac{c^{1-u}}{1-u} - yc \right\}, \quad (29)$$

$$S(u, y) = \Psi(u, y) + .01y - .00005 - .004 \log(.5) + .0008(u - 2)^2. \quad (30)$$

On $u \in [1.5, 2.6]$, $y \in [.2, 12]$, one has $-8 < S < 0$. Extend S to a bounded function $\tilde{S} \in [-8, 0]$ by clipping its two arguments at these verification faces. Let

$$F_R(h, u, y) = \mathbb{E}_{u,y} \int_0^h e^{-.04r} \tilde{S}(U_r, Y_r) dr, \quad (31)$$

where $dU_r = .05 dZ_r^u$ and Y follows (28). Thus $-8h \leq F_R \leq 0$.

THEOREM 5—A restricted-economy upper bound: *Let Z_R be the sum of the following nonnegative localization potentials:*

$$Z_{R,u} = 8h \{ e^{-200(u-1.5)+50h} + e^{-240(2.6-u)+72h} \}, \quad (32)$$

$$Z_{R,y} = 8h \{ e^{-22 \log(y/.2)+22.33h} + e^{-22 \log(12/y)+22.33h} \}. \quad (33)$$

Then

$$V_R(0, 2, 1.25) \leq W_0(2, 1.25, 1.55) + F_R(1, 2, 1.55) + Z_R(1, 2, 1.55). \quad (34)$$

TABLE IV
REFINING THE NO-ADJUSTMENT DUAL WITNESS

Weighted boxes	Dual witness-value enclosure	Width	Seconds
16384	$[-1.33876765, -1.33309541]$	0.00567	2.67
131072	$[-1.33676857, -1.33544680]$	0.00132	15.13
1048576	$[-1.33631272, -1.33599309]$	0.00032	110.24

These intervals enclose the *dual witness value*, not the restricted optimum. Only their upper endpoints are used as upper bounds for that optimum. A separately evaluated restricted policy gives the primal lower bound. Original stopping and all adaptive consumption and portfolio opportunities are retained.

TABLE V
CERTIFIED CURRENT-POLICY EXPENDITURES AND OPTIMAL VALUE OF ACCESS

k	Policy adjustment budget	Optimal access-value interval	Regret target
0.5	$[0.01757696, 0.01757697]$	$[0.06332184, 0.08224255]$	$< .01$
2	$[0.01241866, 0.01241867]$	$[0.04096859, 0.05973218]$	$< .01$
8	$[0.00192681, 0.00192682]$	$[0.00781706, 0.02683234]$	$< .01$

The expenditure column evaluates the displayed learned policy. The access columns bound the difference between the two *optimal* values. The lower endpoint pairs a feasible flexible policy with the restricted dual upper; the upper endpoint pairs the flexible upper with a feasible restricted policy. All endpoints are rounded outward from full-precision records. The cost-two access interval has width below its entire positive gain; at cost eight, the interval still resolves the sign but not a comparably precise magnitude.

The upper bound covers all adaptive original consumption and portfolio controls with $\theta = 0$. Consequently, for any feasible flexible policy with certified payoff lower bound $\underline{J}_k$ and any upper bound $\bar{V}_R$ from (34),

$$V_k(0, 2, 1.25) - V_R(0, 2, 1.25) \geq \max\{0, \underline{J}_k - \bar{V}_R\}. \quad (35)$$

The expectation in (31) has an explicit two-Gaussian representation. We transform time by $r = v^2$, evaluate constant and linear Taylor terms with interval-enclosed weighted moments, and bound the Hessian remainder on each box. Exponential Taylor remainders, Gaussian tails, and localization are included. This produces the executed refinement in Table IV; it does not appeal to a mesh difference as an error estimate. The final restricted-optimum upper bound is -1.33599309 after outward display rounding. The evaluated restricted policy has payoff in $[-1.34503942, -1.34503585]$, so the restricted problem itself has an initial optimal-value enclosure narrower than .00905.

The welfare lower bounds remain strictly positive at all three prices. At $k = 2$ the optimal access gain lies in $[.04096859, .05973218]$; its width is smaller than the lower endpoint. Table VI separately compares the flexible optima at different adjustment prices. The $2 \rightarrow 8$ and $.5 \rightarrow 8$ effects exceed their entire interval widths. The $.5 \rightarrow 2$ comparison resolves the sign but has a larger relative uncertainty. No comparison assumes that a displayed policy is an optimizer or interprets the value-ordering theorem as a pointwise ordering of adjustment controls.

6. A CERTIFIED POLICY RULE ON A CONTINUOUS COST INTERVAL

The preceding value and expenditure enclosures can answer a stronger economic question than evaluation at three selected costs: can one supply an approximately optimal policy for *every* adjustment price in an interval? Repeating training on a fine parameter grid is not enough. An

TABLE VI
ACCURACY-CALIBRATED OPTIMAL WELFARE LOSSES FROM HIGHER ADJUSTMENT COSTS

Cost change	Optimal welfare loss	Interval width	Width below loss
0.5 $\rightarrow$ 2	[0.01263598, 0.03222764]	0.01959166	Sign resolved
2 $\rightarrow$ 8	[0.02318257, 0.04286880]	0.01968623	Yes
0.5 $\rightarrow$ 8	[0.04553582, 0.06537918]	0.01984335	Yes

Each interval encloses $V_{k_1} - V_{k_2}$ in the unchanged original economy. Bounds use independent value endpoints, not fitted derivatives in k . The last column compares the width with the lower endpoint of the effect interval.

unexamined price may lie between grid points, and a policy fitted at one price need not remain approximately optimal at another. The affine dependence of a fixed policy's payoff on the cost coefficient makes a rigorous continuation possible.

Fix $(t, u, x) = (0, 2, 1.25)$. For ordered prices $k_0 < \dots < k_m$, let π_j be a frozen feasible policy and suppose independently computed quantities satisfy

$$L_j \leq J_{k_j}(\pi_j) \leq V(k_j) \leq U_j, \quad b_j^- \leq B(\pi_j) \leq b_j^+. \quad (36)$$

Here B includes the actual random first exit. It is not an undiscounted or unstopped training proxy. Define

$$\ell_j(k) = L_j - (k - k_j) \begin{cases} b_j^+, & k \geq k_j, \\ b_j^-, & k < k_j, \end{cases} \quad \ell(k) = \max_{0 \leq j \leq m} \ell_j(k), \quad (37)$$

$$u(k) = \frac{k_{i+1} - k}{k_{i+1} - k_i} U_i + \frac{k - k_i}{k_{i+1} - k_i} U_{i+1}, \quad k \in [k_i, k_{i+1}]. \quad (38)$$

Using the upper expenditure endpoint on *both* sides of a node would be wrong: a lower cost rewards the policy's expenditure, so its lower endpoint must be used on the left.

THEOREM 6—Certified parameter continuation: *Suppose the policy set, state dynamics, and stopping contract do not depend on k , and (36) holds. Select the lowest-index maximizer $j(k)$ of (37). For every real $k \in [k_0, k_m]$,*

$$\ell(k) \leq J_k(\pi_{j(k)}) \leq V(k) \leq u(k), \quad 0 \leq V(k) - J_k(\pi_{j(k)}) \leq u(k) - \ell(k). \quad (39)$$

On each node interval the maximum of $u - \ell$ occurs at an endpoint or a breakpoint of the upper envelope of the affine functions ℓ_j . If the inputs are rational, these candidate points and gaps admit an exact finite computation. Strictly bounding all these gaps by ε certifies (39) on the entire continuous price interval, without a parameter-grid approximation error.

The proof in Appendix A uses the affine fixed-policy payoff and convexity of the optimal value. It does not require differentiability of V , an attained optimal control, or a convergence theorem for the network optimizer. This use of convexity is distinct from assuming convexity of the shared-parameter training objective.

6.1. Construction and stopping rule

The starting nodes are the three independently certified prices .5, 2, and 8. A disclosed pilot adds 4. Thereafter a failed node interval is bisected, in increasing-price order; each new price

TABLE VII
CONSTRUCTIVE CONTINUATION TO THE UNIFORM ECONOMIC TOLERANCE

Certified prices	Uniform regret upper	Task-seconds	Target met
3	0.023989184	0.00	No
4	0.014155490	20.38	No
7	0.011507236	85.20	No
13	0.010387647	219.98	No
17	0.010002257	304.81	No
18	0.009994025	325.31	Yes

The maximum covers every real price in $[.5, 8]$, at the fixed initial state. Task-seconds sum successful new price-case durations, including polishing and verification; parallel tasks overlap. They exclude the inherited three certificates and interrupted attempts and are not end-to-end CPU time. All interruption records are preserved separately. The acceptance criterion is the exact rational gap, not the displayed runtime.

receives its own feasible actor, exact-first-exit payoff enclosure, expenditure enclosure, and affine-preference dual upper bound. Training uses the nearest inherited actor only as a warm start. Output polishing and dual fitting use the preceding deterministic routines with explicitly audited input/output substitutions. They do not use an optimal-value label. The original source files are unchanged.

All final policy evaluations use 16,384 outward time rectangles, and each optimal-value upper uses 65,536 weighted source boxes. Artificial verification faces carry the same explicit localization allowances as in Theorem 3; none becomes an economic stopping boundary. Each continuous-price test then converts the binary64 certificate endpoints to their exact rational values. A monotone upper-hull algorithm constructs all relevant line intersections. No numerical mesh, random evaluation price, interpolation residual estimate, or optimization tolerance replaces this last test.

The final library contains eighteen nodes and fifteen newly certified policies. It yields

$$\sup_{k \in [.5, 8]} \{V(k) - J_k(\pi_{j(k)})\} \leq .009994024846927508 < .01. \quad (40)$$

The displayed upper is rounded outward from the exact rational maximum. The largest gap occurs at an envelope intersection near $k = 7.7422$, not at a training price. Checking only the eighteen training prices would therefore miss the relevant acceptance calculation. This is also a reason to retain all failed refinement stages rather than show only the last table row.

PROPOSITION 3—Finite refinement under local slack: *Suppose each newly supplied node certificate satisfies $U_j - L_j \leq \varepsilon - \eta$ for a common $\eta > 0$, and $0 \leq b_j^- \leq b_j^+ \leq \bar{B}$. On an interval of length h ,*

$$\max(u - \ell) \leq \max\{U_i - L_i, U_{i+1} - L_{i+1}\} + \bar{B}h/4. \quad (41)$$

Consequently, exact bisection terminates after finitely many refinements when all node intervals have length less than $4\eta/\bar{B}$.

The proposition quantifies the continuation step, not the ability of an arbitrary optimizer to supply the stipulated node slack. A fixed witness class can have a structural accuracy floor. The executed result (40) does not rely on an unproved uniform-slack claim: its actual finite library passes the exact test. The tighter upper-hull test, rather than the conservative bound (41), is used for acceptance.

6.2. Economic interpretation between certified prices

The library delivers a policy for a queried cost without retraining. It selects one frozen sixteen-slab control rule for the whole horizon and applies the original liquidation contract if exit occurs. It does not switch policies according to an unverified neural value between states. The guarantee is against the full adaptive optimum, not merely against the best policy in the library.

For prices $p < q$, the continuation bounds imply

$$V(p) - V(q) \in [\ell(p) - u(q), u(p) - \ell(q)] \cap [0, \bar{B}(q - p)]. \quad (42)$$

Similarly, the value of access lies in $[\ell(k) - \bar{V}_R, u(k) - \underline{V}_R]$. The original certified access gain at $k = 8$ and monotonicity imply a positive access gain throughout $[.5, 8]$. This assertion resolves the sign, not a small relative uncertainty for every price. The sharper effect-to-width comparisons at $k = 2$ and for the $2 \rightarrow 8$ experiment remain those already reported above.

If an optimizer exists at an interior price k , all available secants may be intersected to obtain

$$B(\pi_k^*) \geq \max \left\{ 0, \sup_{q > k} \frac{\ell(k) - u(q)}{q - k} \right\}, \quad (43)$$

$$B(\pi_k^*) \leq \min \left\{ \bar{B}, \inf_{p < k} \frac{u(p) - \ell(k)}{k - p} \right\}. \quad (44)$$

Using a finite subset of these secants remains valid and avoids numerical differentiation of a possibly nonsmooth value function. The deployed policies' independently enclosed expenditures remain separate objects; no optimal adjustment budget is inferred from them without the requisite optimal-value comparison.

The new object is the executable parameter-uniform policy certificate and its exact stopping rule. Neither convexity of a supremum of affine functions nor classical policy verification is claimed as new. In combination with the original stopped dual, the construction closes an additional economically meaningful computational chain: a continuum of cost counterfactuals, an implementable policy at every cost, and an explicit absolute error smaller than the declared tolerance. It does not convert the separate external fixed-budget experiments into a universal performance claim.

7. GLOBAL ACTION CERTIFICATION WITHOUT A TENSOR GRID

Consider the following minimization problem, which is also used for the external comparison:

$$dX_t = 2a_t dt + \sqrt{.32} dW_t, \quad X_0 = 0, \quad a_t \in [-1, 1]^d, \quad (45)$$

$$f_d(a) = \frac{1}{d} \sum_{i=1}^d \{a_i^2 + .1 \sin^2 a_i\} + .3 \sin \left(\sum_i a_i \right), \quad (46)$$

$$q_d(x) = \frac{1}{d} \sum_{i=1}^d (x_i - .5)^2 + .2 \cos \left(d^{-1/2} \sum_i x_i \right), \quad (47)$$

$$V_d^* = \inf_a \mathbb{E} \left[\int_0^1 f_d(a_t) dt + q_d(X_1) \right].$$

The objective has genuinely coupled nonconvex actions. No analytic optimal-action or optimal-value labels are supplied to training. At a fixed critic gradient p , the action-dependent Hamilto-

TABLE VIII
EXECUTED CONTINUOUS-ACTION CERTIFICATES

Action dim.	Tolerance	Verified gap	Scalar boxes	Convex subproblems	Seconds
8	1e-03	0.000258	33	264	0.299
8	1e-05	6.72e-06	47	376	0.493
16	1e-03	0.000931	37	592	0.476
16	1e-05	4.42e-06	57	912	0.751
32	1e-03	0.00068	85	2720	1.498
32	1e-05	3.52e-06	103	3296	1.991
64	1e-03	0.000647	163	10432	6.041
64	1e-05	3.06e-06	185	11840	5.955
128	1e-03	0.000948	317	40576	19.229
128	1e-05	4.68e-06	339	43392	20.651

One frozen jet is used per dimension, at both tolerances. At $d = 8, 16$, these are gradients of retained learned critics at recorded states; displayed gaps include interval neural-derivative allowances. At larger dimensions they are independently generated jets for the same Hamiltonian family. The experiment measures certificate work for these instances, not an average-case scaling law or a whole-state neural PDE residual.

nian is

$$H_p(a) = \sum_i f_i(a_i) + .3 \sin z, \quad f_i(s) = \frac{s^2 + .1 \sin^2 s}{d} + 2p_i s, \quad z = \sum_i a_i.$$

Each f_i is strongly convex with modulus at least $m = 1.8/d$.

THEOREM 7—Scalar global partition and convex dual bounds: *For any multiplier λ and $z \in [-d, d]$, put*

$$D(\lambda) = \sum_i \min_{|s| \leq 1} \{f_i(s) - \lambda s\}.$$

On a scalar interval $z \in [z_0 - r, z_0 + r]$, a lower bound for the full Hamiltonian is

$$D(\lambda) + \lambda z_0 + .3 \sin z_0 - |\lambda + .3 \cos z_0| r - .15 r^2. \quad (48)$$

Any feasible action supplies an upper bound. Each inner convex minimum can itself be bounded below by $q(s_0) - e(s_0)^2/(2m)$, where $q = f_i - \lambda s$ and e is the derivative magnitude in a feasible descent direction, or the full derivative magnitude at an interior point. Outward versions of these bounds, followed by scalar subdivision and pruning, give a global gap on $[-1, 1]^d$.

If a stored gradient $\hat{p}$ differs from the true neural gradient by componentwise bounds δ_i , the feasible action's true-gradient gap is at most its stored-gradient gap plus $4 \sum_i \delta_i$.

Floating-point root finding only proposes multipliers and actions. It is not trusted to solve the convex subproblems exactly. Even an inaccurate proposal remains valid after the strong-convexity allowance. This structure removes tensor enumeration; it does not establish a dimension-free complexity theorem or solve a generic nonseparable nonconvex Hamiltonian.

The 128-dimensional 10^{-5} calculation uses 339 scalar boxes and 43,392 scalar convex subproblems. It replaces neither raw actor training nor the preceding finite tensor safeguard in retrospect. At the learned eight- and sixteen-dimensional jets, interval forward differentiation of the stored critic weights bounds the ℓ^1 gradient discrepancies below 4.584×10^{-17} and

TABLE IX

ABSOLUTE LOWER BOUNDS FOR THE ORIGINAL NONCONVEX CONTROL PROBLEM

d	Quadrature cells	Relaxed-value enclosure	Original-value lower	Seconds
8	1024	$[-0.08599082, -0.08384626]$	-0.08599082	0.260
8	4096	$[-0.08518701, -0.08465086]$	-0.08518701	1.081
8	16384	$[-0.08498598, -0.08485194]$	-0.08498598	4.506
16	1024	$[-0.20520651, -0.20360335]$	-0.20520651	0.271
16	4096	$[-0.20460665, -0.20420585]$	-0.20460665	1.067
16	16384	$[-0.20445644, -0.20435623]$	-0.20445644	4.212
32	1024	$[-0.28757809, -0.28684340]$	-0.28757809	0.245
32	4096	$[-0.28730394, -0.28712026]$	-0.28730394	0.988
32	16384	$[-0.28723515, -0.28718923]$	-0.28723515	4.172

A row encloses the relaxed value. Its lower endpoint is also a lower bound for the original bounded-control value. The width shown by successive enclosures is quadrature uncertainty; it does not include or estimate the structural relaxation gap.

7.877×10^{-17} , respectively. These allowances are propagated into the action gap. Theorem 2 identifies the additional state-cover and residual quantities that would be needed for a whole-policy guarantee.

8. AN ABSOLUTE BOUND AND A FROZEN EXTERNAL COMPARISON

8.1. A rigorous lower bound for the unchanged nonconvex family

Let $r_0 = 1 + .1 \sin^2(1)$, $\sigma^2 = .32$, and $\lambda_d = \sigma^2 r_0 / (2d)$. Since $\sin^2 a \geq \sin^2(1) a^2$ for $|a| \leq 1$, replacing the running cost by $r_0 \|a\|^2 / d - .3$ and allowing unbounded controls gives a relaxation. The original terminal function q_d is retained.

PROPOSITION 4—Scalar-integral absolute lower bound: *Let Z be normal with mean $2\sqrt{d}/(r_0 + 4)$ and variance $\sigma^2 r_0 / (r_0 + 4)$. Then*

$$V_d^* \geq \frac{.25r_0}{r_0 + 4} + \frac{\sigma^2 r_0}{4} \log(1 + 4/r_0) - .3 - \lambda_d \log \mathbb{E} \exp \left[-\frac{.2}{\lambda_d} \cos Z \right]. \quad (49)$$

An outward enclosure of the positive expectation yields a rigorous numerical lower bound for the original objective (47).

The relaxation is an upper/lower-bound device, not a new training problem with a known manufactured value. The terminal cosine, which couples all state coordinates, remains in the scalar expectation. Its tails are bounded analytically and its interior is integrated by interval rectangles.

8.2. Holdout design and exact deployment interpretation

The comparison retains the original actor and critic architectures: two tanh hidden layers with 48 units each, a bounded actor output, and the same terminal-value embedding. NBO uses separated evaluation and improvement with Adam, three critic steps per actor step, 256 training states, and four successors. SOC-MartNet uses its pinned author solver, RMSprop, the 600-output test network, and the inherited $J = 2$, $K = 1$ multiplier setup. These different optimization systems are disclosed rather than described as identical implementations.

The committed protocol freezes dimensions 8, 16, 32 and training budgets of 10 and 30 seconds. At each dimension, both methods receive the same six validation trials: two seeds

TABLE X
FROZEN HOLDOUT COMPARISON: TWELVE SEED PAIRS PER PANEL

d	Budget	NBO cost	SOC cost	LQ cost	Median difference	Adjusted p
8	10	0.5812	1.0842	0.5121	-0.4783	0.001465
8	30	0.5785	1.0557	0.5121	-0.4348	0.001465
16	10	0.4663	0.7893	0.1740	-0.1672	1
16	30	0.4630	0.8456	0.1740	-0.2878	.438
32	10	0.3813	1.2011	0.0854	-0.8691	0.001465
32	30	0.3799	1.0218	0.0854	-0.6962	0.001465

Smaller realized cost is better. The median difference is NBO minus SOC-MartNet. The last column is a one-sided exact sign-test p -value adjusted across the six predeclared dimension–budget panels. The sign test concerns the joint training-and-audit pipeline under independent seed draws, not an outward numerical error guarantee.

TABLE XI
ABSOLUTE REFERENCE EXCESS AND ACTUAL TRAINING CLOCKS

d	Budget	Lower bound	NBO excess	SOC excess	NBO sec.	SOC sec.
8	10	-0.08499	0.6662	1.1692	10.002	10.006
8	30	-0.08499	0.6635	1.1407	30.004	30.008
16	10	-0.20446	0.6707	0.9938	10.002	10.005
16	30	-0.20446	0.6674	1.0500	30.004	30.006
32	10	-0.28724	0.6686	1.4883	10.003	10.007
32	30	-0.28724	0.6672	1.3090	30.003	30.007

Excess is the estimated mean policy cost minus the rigorous original-value lower bound. It provides an absolute reference axis but is not itself a deterministic regret enclosure: sampling error and floating-point rollout evaluation are not outward certified. Training clocks include completed-update overshoot and exclude separately recorded warmup and setup.

and three learning-rate multipliers. Selection minimizes mean independent validation cost, with a fixed tie rule. Twelve different holdout seeds, 720–731, are then evaluated at each dimension–budget pair. Within each pair, actor and critic starting weights are identical and their hashes are checked. Framework warmup precedes both clocks, and method order is seed-randomized. Training-only and setup-plus-training clocks are retained. The baseline $a_i(t, x) = \text{proj}_{[-1,1]}[-2(x_i - .5)/\{1 + 4(1 - t)\}]$ is specified independently from the associated quadratic problem and is evaluated on the unchanged nonquadratic objective.

Each trained network is deployed as a sample-and-hold control on 128 dates and evaluated with 4,096 independent, paired Gaussian paths. In (47), drift under a held action and volatility are constant on a slab; the Gaussian transition and running-cost integral are therefore exact in real arithmetic for *that deployed continuous-time policy*. A 64-date evaluation uses aggregated common increments and changes the deployment policy. Its difference from the 128-date result is not called an Euler-error bound. Floating-point network evaluation and Monte Carlo uncertainty remain distinct from the outward deterministic lower bound in Table IX.

The holdout results favor NBO over the author implementation in mean cost in all six panels, but their interpretation is not uniform. In dimensions 8 and 32, NBO has lower paired cost for all twelve seeds at both budgets; each multiplicity-adjusted sign-test value is .001465. In dimension 16 the adjusted values are 1 and .438, so these panels do not establish a robust seed-level ordering under the declared test. Tripling the budget gives only small improvements in NBO mean cost, from .5812 to .5785, .4663 to .4630, and .3813 to .3799 respectively. Most importantly, the independently specified quadratic-feedback baseline has lower mean cost than both learned methods in every panel. Its mean costs are .5121, .1740, and .08544 in dimensions 8, 16, and 32. These retained results support the stated NBO–SOC-MartNet comparison at the

frozen budgets, not a claim that NBO is the best available control rule or that additional training has already reached an accurate optimum.

The raw records retain every seed difference, the median and interquartile range, approximate Student intervals, costs for all three rules, path standard errors, training curves, optimizer clocks, and checkpoints. Approximate Student intervals are descriptive; the exact sign calculation avoids assuming Gaussian seed-level differences. The source protocol predates holdout execution, while the benchmark family and adapters themselves were developed in earlier revisions. Holdout seeds do not turn one problem family into evidence of general superiority across economic environments. Two budget points are not a matched-accuracy frontier, and a finite absolute lower bound need not be tight enough to rank methods at a small prescribed regret.

9. METHOD-SPECIFIC TUNING AND AN INDEPENDENT REPLICATION

The preceding holdout answers a precisely frozen comparison, but its scalar learning-rate search leaves an important design question: does the ordering survive a more consequential method-specific tuning effort? We answer this question with a new experiment, rather than choosing new hyperparameters on the already observed holdout or suppressing the earlier results. The objective, diffusion, action box, horizon, equal actor/critic initializations, author solver commit, and 128-date deployed policy are unchanged. The new protocol fixes tuning seeds 1600 and 1601 and holdout seeds 1820–1831 before execution. None of these is an R9 holdout seed.

9.1. *Prospectively specified equal tuning opportunities*

For each dimension and method, twelve initial configurations and two prospectively specified extension configurations are each trained for five nominal seconds on two tuning seeds. Thus each method receives 140 nominal training seconds per dimension. Completed-update overshoot, initialization, validation, and holdout clocks are recorded separately. The methods receive equal nominal training opportunity, not equal numbers of gradient operations or an assertion that their entire hyperparameter spaces have been optimized.

The shared learning-rate multiplier set expands from $\{1/3, 1, 3\}$ to $\{1/3, 1, 3, 9, 27\}$. NBO additionally varies evaluation updates per actor improvement and transition particles. SOC-MartNet additionally varies the adversary learning rate, the number of sinusoidal test outputs, the solver’s J/K parameters, and the initial and updating multiplier parameters. The complete twelve-configuration lists appear in the frozen executable protocol; unchanged settings and unsuccessful candidates are not omitted. The pinned author training implementation is unmodified. Method order is randomized within tuning seed and candidate, and actor and critic initial hashes must agree within each holdout pair.

After the first twelve configurations, selection minimizes the mean validation cost over the two tuning seeds, with an index-based tie rule. A selected multiplier at the current high or low boundary is extended by a factor of three in that direction. If the multiplier is interior, the additional slot varies transition-particle count for NBO or adversary rate for SOC. Selection is repeated before the second extension. These rules are specified before execution and do not inspect the new holdout. Failed tuning trials receive infinite selection score while their raw records are preserved. Validation uses 1,024 common Brownian paths per tuning seed, and holdout evaluation uses 4,096 new common paths per seed. All recorded outcomes in this execution are finite, and all 168 tuning trials and 36 holdout pairs are retained.

All six selected learning-rate multipliers are interior to the enlarged tested multiplier set. At $d = 8$, NBO selects $m = 9$, while SOC selects $m = 3$ with initial multiplier and multiplier step both 30. At $d = 16$, NBO selects six evaluation steps and sixteen transition particles with $m = 3$;

TABLE XII
SELECTED CONFIGURATIONS AND ACTUAL TUNING-TRAINING OPPORTUNITY

d	NBO selected	SOC selected	NBO sec.	SOC sec.
8	3 ($m = 9$)	11 ($m = 3$)	140.15	140.29
16	12 ($m = 3$)	2 ($m = 3$)	140.09	140.18
32	1 ($m = 1$)	3 ($m = 9$)	140.17	140.31

Configuration indices are zero based and bind to the complete frozen lists and extension records. m is the actor/critic learning-rate multiplier. Actual seconds sum all 28 tuning fits per method and dimension, including completed-update overshoot but excluding separately retained validation and setup.

TABLE XIII
NEW HOLDOUT AFTER EQUAL-BUDGET METHOD-SPECIFIC TUNING

d	NBO	SOC	LQ	Mean diff.	Median diff.	Wins	Adj. p
8	0.5542	0.7928	0.5125	-0.2386	-0.1967	11/12	0.00952
16	0.4547	0.8067	0.1741	-0.3520	-0.1212	8/12	0.58154
32	0.3824	0.4586	0.0852	-0.0762	-0.0427	7/12	1.00000

Smaller realized cost is better. Differences are NBO minus SOC, using paired audit paths and twelve independently seeded training-and-audit pipelines. The last column applies the prospectively fixed three-panel sign-test correction. Costs are sampled policy evaluations, not outward deterministic certificates. The LQ column retains the independently specified clipped feedback under the unchanged nonconvex objective.

SOC selects its default method-specific parameters with $m = 3$. At $d = 32$, NBO selects $m = 1$ and SOC selects $m = 9$, with the respective default remaining parameters. The large learning-rate boundary identified in the earlier report is therefore tested beyond its old limit. Interior selection on this finite set does not establish global tuning optimality or exclude improvements along an unsearched interaction. The experiment tests a finite, disclosed method-specific tuning budget.

9.2. Independent holdout and protocol sensitivity

The selected configurations are fixed before their twelve new holdout pairs are run at a ten-second nominal training budget. The primary test is again the one-sided exact sign test, now with Bonferroni correction across the *three new* predeclared panels. This does not change the six-panel correction of the earlier experiment. Because the historical evidence motivated the new experiment, the new study is a prospective post-review replication, not a retroactive preregistration of the original benchmark design.

The NBO mean differences are $-.23863$, $-.35199$, and $-.07623$ in dimensions 8, 16, and 32. The corresponding paired win counts are 11, 8, and 7 out of twelve, with adjusted sign-test values .00952, .58154, and 1. Only the eight-dimensional panel establishes the declared robust seed-level ordering at level .05. In dimension 16, four seeds again favor SOC. In dimension 32, the expanded tuning yields a substantially smaller mean difference and no significant directional ordering, although the earlier frozen experiment favored NBO in all twelve seeds. The improved SOC result is consistent with its selected $m = 9$, which the old scalar grid excluded; this association is not an isolated causal estimate of that hyperparameter because tuning, validation seeds, and holdout seeds all differ.

The independently specified clipped-feedback policy has lower mean realized cost than either trained pipeline in each new panel, as it did in the previous experiment. Its favorable performance does not make its cost a lower bound for a minimization problem, and it does not prove global

optimality. These outcomes distinguish three questions: whether a configured NBO pipeline improves on a configured external solver, whether the ordering is robust to a changed tuning protocol, and whether either neural policy has small absolute error. The new experiment supplies direct evidence for the second question and retains all evidence relevant to the first; Section 10 supplies a separate exact-reference experiment for the third. Favorable absolute-accuracy results in another model are not used to erase the sixteen- or thirty-two-dimensional sign-test outcomes here.

The evidence pipeline collects with an unconditional completion policy. Every planned cell has explicit status before dependency installation; successful outputs, failed outputs, and missing-cell metadata are versioned before aggregate validity is checked. Separate immutable commits identify the frozen study source and completed raw results. No branch name alone is used as a review target, and the 36 new pairs neither replace nor pool selectively with the 72 earlier pairs.

10. FROM IMPLEMENTED ACTOR ERRORS TO ABSOLUTE-ACCURACY FRONTIERS

The original stopped economy establishes economically resolved optimality bounds. A complementary experiment asks a different question: how much computation is required to attain a common absolute accuracy when the continuous optimum is independently known? This experiment changes neither the economy in Section 2 nor the coupled nonconvex comparison. It adds reference-solvable mechanisms with different drift, noise, and control-cost conditioning. An exact reference is not replaced by the best realized cost of a competing policy.

Consider independent coordinates with

$$\begin{aligned} dX_t^j &= (a_j X_t^j + b_j u_t^j) dt + \sigma_j dW_t^j, \\ J(u) &= \frac{1}{d} \mathbb{E} \left[\int_0^1 \sum_{j=1}^d (q_j (X_t^j)^2 + r_j (u_t^j)^2) dt + \sum_{j=1}^d w_j (X_1^j)^2 \right], \end{aligned} \quad (50)$$

where $q_j \geq 0$, $r_j, w_j > 0$, and $b_j > 0$. Controls are unbounded, adapted, and square integrable. The action constraints and nonconvex running cost of the external comparison are *not* silently imposed on this linear-quadratic problem. Rather, the latter supplies an independently solvable accuracy experiment. The implemented actor observes $X_{i/N}$ and holds $u_t = -K_i X_{i/N}$ until the next decision date. It is a linear output map with time gates, not a continuous-feedback deployment of $-K_i X_t$ and not the width-48 tanh actor used in the external comparison.

PROPOSITION 5—Two independent error decompositions: *Let $p_j(1) = w_j$ and $-\dot{p}_j = q_j + 2a_j p_j - b_j^2 p_j^2 / r_j$. The continuous optimum is*

$$V^* = \frac{1}{d} \sum_j \left[p_j(0) (X_0^j)^2 + \sigma_j^2 \int_0^1 p_j(t) dt \right], \quad (51)$$

and every admissible control satisfies

$$J(u) - V^* = \frac{1}{d} \sum_j \mathbb{E} \int_0^1 r_j \left(u_t^j + \frac{b_j p_j(t)}{r_j} X_t^j \right)^2 dt. \quad (52)$$

For decision interval $h = 1/N$, let J_h^ be the optimum over held-action rules, with Riccati coefficients P_{ij} , optimal gains K_{ij}^* , and positive action curvatures D_{ij} specified in Appendix B.*

For any implemented held-action rule, including a nonlinear neural rule,

$$J(u) - V^* = (J_h^* - V^*) + \frac{1}{d} \sum_{i,j} D_{ij} \mathbb{E} [(u_i^j + K_{ij}^* X_{ih}^j)^2]. \quad (53)$$

Consequently, output-error bounds $\mathbb{E}[(u_i^j + K_{ij}^* X_{ih}^j)^2] \leq \varepsilon_{ij}^2$ give a quantitative optimization allowance $d^{-1} \sum_{i,j} D_{ij} \varepsilon_{ij}^2$, in addition to the separately evaluated deployment error $J_h^* - V^*$. No convergence assumption on Adam or on shared neural parameters is used.

The proposition is a specialization of quadratic verification and Bellman completion of squares, not a new general Riccati theorem. Its role is to connect an *implemented* policy error to an absolute economic objective. The finite-horizon quadratic Bellman construction requires exactly N backward improvements per coordinate. In this output class, evaluation and improvement are closed operations, and the backward recursion solves the held-action problem rather than merely decreasing a stochastic training loss. The alternative Adam implementation optimizes the same gain outputs using an independently coded exact-moment objective. Its errors are assessed after the recorded updates; a universal stochastic-optimization convergence theorem is not inferred.

10.1. Independent reference and deployment verification

Write $c_j = b_j^2/r_j$, $\lambda_j^2 = a_j^2 + c_j q_j$, and $s = 1 - t$. Define

$$A_j(s) = \cosh(\lambda_j s) + \frac{\sinh(\lambda_j s)}{\lambda_j} (-a_j + c_j w_j),$$

$$B_j(s) = w_j \cosh(\lambda_j s) + \frac{\sinh(\lambda_j s)}{\lambda_j} (q_j + a_j w_j).$$

The continuous Riccati solution is $p_j(t) = B_j(1 - t)/A_j(1 - t)$, with the continuous limit when $\lambda_j = 0$. Its integral equals $(\log A_j(1) + a_j)/c_j$. Thus the reference needs no time-stepping solver or simulated paths. Sixty-decimal interval arithmetic encloses the exponentials, logarithms, and divisions. Every coefficient in the certificate is the exact binary64 number stored in the instance file, and every gain is the exact stored policy output. Exported interval endpoints are enlarged by one binary64 unit in the outward direction.

A different calculation evaluates the original continuous cost of each held-action policy. Conditional on the state at a decision date,

$$X_{ih+s}^j = \left(e^{a_j s} - b_j K_{ij} \int_0^s e^{a_j v} dv \right) X_{ih}^j + \sigma_j \int_0^s e^{a_j(s-v)} dW_{ih+v}^j.$$

Exact second-moment propagation and the integrated running cost therefore determine $J(u)$. Appendix B gives all coefficients and the verification proof. This is not an Euler approximation, and a difference between two meshes is not used as an error estimate. In particular, the reported interval $[\underline{J} - \overline{V}, \overline{J} - \underline{V}]$ encloses absolute regret in (50), including deployment error. Stored policies, model coefficients, source hashes, and the two independent evaluators are supplied in the replication package.

10.2. Reference-solvable mechanisms and executed accuracy

The prospective protocol specifies three cases. Tracking has $d = 8$, $a = 0$, $b = 2$, $q = 0$, $r = w = 1$, initial centered state $-.5$, and $\sigma = .5656854249492381$. The mean-reverting case

TABLE XIV
ABSOLUTE CONTINUOUS-MODEL REGRET: OUTWARD UPPER ENDPOINTS

Case	B: $N = 16$	B: $N = 32$	B: $N = 128$	Adam (400)
Tracking	0.00839748	0.00409984	0.00100625	0.00409984
Mean-reverting	0.00429055	0.00211098	0.00052088	0.00211101
Ill-conditioned	0.01421242	0.00699643	0.00172851	0.00699643

Each entry is rounded upward from an interval certificate of the original continuous cost of the stored actor. Columns B denote quadratic Bellman construction at the stated decision count. Adam uses 32 decisions and 400 updates. These are structured-policy accuracy results, not a substitution for the external neural comparison.

TABLE XV
FIRST CERTIFIED ATTAINMENT ON THE RECORDED COMPUTATION PATHS

Case	Method	.01 (sec.)	.003 (sec.)	.001 (sec.)
Tracking	B	0.023	0.082	—
Tracking	A	0.626	—	—
Mean-reverting	B	0.015	0.053	0.189
Mean-reverting	A	0.641	2.380	—
Ill-conditioned	B	0.109	0.395	—
Ill-conditioned	A	0.741	—	—

Time includes all earlier generation and certificate checks on the indicated path, but excludes dependency installation and one-time optimizer construction. B is the quadratic Bellman constructor; A is Adam on gain outputs. A dash means the target was not attained among the predeclared checkpoints. Timing is a single-machine descriptive measurement, not a confidence interval or a hardware-independent complexity comparison.

has $d = 8$, $a = -1$, $b = q = w = 1$, $r = .25$, $\sigma = .4$, and initial state one. The ill-conditioned case has $d = 16$ and linearly spaced $a \in [-1, .5]$, $b \in [.5, 2]$, $q \in [.1, 2]$, $\sigma \in [.15, .65]$, $w \in [.5, 2]$, and initial states in $[-1, 1]$, with geometrically spaced $r \in [.1, 4]$. The last case contains both stable and unstable uncontrolled coordinates. Exact stored coefficients, rather than rounded endpoints in this description, define each computation.

The quadratic Bellman constructor is evaluated at $N = 4, 8, 16, 32, 64, 128$. Adam uses $N = 32$, initially zero gains, learning rate .05, standard moment coefficients (.9, .999), and checkpoints at 0, 25, 100, and 400 updates. The deterministic optimizer configuration is recorded with the executed instances; no tuning search is performed for this reference suite. No holdout claim is attached to these analytical test cases. All checkpoints are retained, including those that miss a stated accuracy threshold.

Table XIV shows that refinement reduces deployment error in every case. At 128 decisions, all three absolute regret upper bounds are below .003. At 32 decisions, all three are below .01. Adam approaches the same 32-decision optimum: increasing its updates cannot eliminate the deployment floor at that fixed decision count. This distinction is observed directly, not inferred from a nearly flat training curve. The .001 target is attained on the planned mesh sequence for the mean-reverting case, but not for tracking or the ill-conditioned case; these outcomes remain in the package rather than being removed from the frontier.

The accuracy axis is the same original-cost regret for both policy generators within each case. The computation axis includes rejected earlier meshes or checkpoints rather than reporting only the final inexpensive verification. The exact quadratic constructor uses analytic model structure that the general tanh and SOC implementations do not exploit; its faster times do not establish a neural-versus-SOC efficiency ranking. Together with the unchanged stopped economy and coupled nonconvex family, the added cases test first-exit verification, adapted

adjustment, nonconvex coupling, mean reversion, and differing control-cost conditioning. They do not supply new recursive-utility or game experiments. Those extensions retain their explicit assumptions and complete historical material.

11. RELATION TO RIGOROUS CONTROL AND NEURAL COMPUTATION

The strongest claims should be compared with methods that establish bounds, not only with methods that train networks. Theorem 1 is an application of classical stopped verification to smooth upper and lower witnesses. Its role here is to handle the actual liquidation trace without presupposing a smooth or boundary-regular optimal value. Theorem 2 supplies a computation-facing cover, derivative, and oracle allowance. Neither theorem claims that classical verification was absent from the literature.

Monotone, stable, consistent approximations with comparison provide convergence to viscosity solutions in the framework of Barles and Souganidis (1991). Controlled Markov-chain approximation is developed systematically by Kushner and Dupuis (2001); adaptive sparse-grid methods address high-dimensional dynamic models in Brumm and Scheidegger (2017). Such results concern specified approximation schemes and their hypotheses. The finite telescoping budget in the preserved manuscript names the operator discrepancies needed for a continuous statement; an observed mesh difference does not supply them. The original-payoff certificates here instead evaluate continuous generators or exact Gaussian moments directly, with separately bounded analytic and arithmetic remainders. They do not assert a convergence rate for the inherited killed-Euler scheme.

Primal–dual numerical control methods also pair feasible decisions with bounds. Brown, Smith, and Sun (2010) relax nonanticipativity with penalties and establish weak and strong duality for stochastic dynamic programs. Brown and Smith (2011) use related bounds in portfolio optimization with transaction costs. Our restricted and flexible constructions are market-deflator witnesses with explicit stopped localization; they are not claims that policy-plus-upper-bound analysis is new. Theorem 3 adds a verified affine-preference source and a conditional-coefficient variance allowance. Its distinction from an information-relaxation bound with an uncomputed penalty is that every source integral, variance constant, and face allowance is evaluated here. Lasserre, Henrion, Prieur, and Trélat (2008) develop occupation-measure and moment relaxations for polynomial nonlinear optimal control under their stated assumptions. We do not invoke their deterministic convergence theorem for the present logarithmic, stochastic, controlled-exit problem. The explicit source bounds, interval polynomial conversions, Gaussian tails, and trace inequalities are verified here instead.

Theorem 3, Proposition 6, and Theorem 5 identify the current and preceding model-specific constructions behind the economic bounds. The affine-preference theorem retains costly adjustment through an adapted conditional coefficient, bounds its randomness quantitatively, and closes the declared central-state target. The fitted generator proposition couples a polynomial witness to a small localization potential; the restricted theorem cancels every portfolio term before computing a bounded-source expectation. Theorem 7 exploits rank-one action coupling and strong convexity of the remaining scalar problems. It is a structured convex-duality and branch-and-bound construction, not a generic replacement for global nonconvex optimization. Proposition 4 preserves the nonquadratic terminal payoff in a computable relaxation of the exact comparison problem.

Neural policy iteration and neural PDE solvers are important computational antecedents. Jacka and Mijatović (2015) analyze continuous-time policy improvement. Kim, Cho, Kim, and Kim (2026) separate iteration, policy-network, and residual errors in entropy-regularized control; Kim, Kim, Kim, and Cho (2026) study interior bounds and the mismatch between

collocation and closed-loop performance. Lee and Kim (2025) combine policy iteration and operator learning. Cai, Fang, and Zhou (2025) impose martingale conditions without an explicit Hamiltonian infimum. NBO’s raw training comparison is with a pinned implementation of the last method, not a locally recreated approximation described as author code. Our certificates can also audit a policy produced by another solver. This portability is a property of independent verification, while the training results remain evidence about the specifically recorded NBO and SOC-MartNet adapters.

Recursive utility, temporal-self equilibria, and games require their own evaluation operators and optimality notions. Their complete equations, assumptions, finite gain bounds, failures, and earlier numerical results are preserved in the supplement. No new large-scale recursive or game accuracy result is inferred from the additive-utility certificates above. The stochastic-trace bias identities and local-optimizer counterexamples are preserved for the same reason: they identify conditions that a neural training loss cannot replace.

12. CONCLUSION

The original continuous preference-adjustment economy now has certified central-state policies at the declared .01 regret tolerance for all three cost coefficients. A feasible learned-and-polished policy supplies each lower bound, while an independent affine-preference dual supplies an upper bound over the original continuous adaptive controls. This closes the numerical target without deleting the original stopping contract or replacing it by a manufactured-payoff problem.

The cost-continuation construction extends this guarantee from three isolated prices to every real price between .5 and 8. Its eighteen-policy library has a largest rigorously enclosed regret of .00999403. Only a finite set of exact envelope breakpoints is checked, while convexity and fixed-policy cost transfer establish the continuum statement. The policy is selected once from the library for the given price; the guarantee remains a central-initial-state result, not an uncomputed whole-state neural certificate.

The value of access and the welfare cost of more expensive adjustment are resolved at a precision calibrated to the reported effects. At coefficient two the optimal access gain is at least .04096859; raising the coefficient to eight lowers optimal welfare by at least .02318257. Both guarantees exceed their corresponding interval widths. Optimal adjustment budgets obey the analytical ordering and a valid secant enclosure, but are not equated with the budgets of approximate policies.

The constructive tools also retain their distinct roles: fitted Bernstein witnesses, stopped Gaussian-moment evaluation, market-deflator localization, quantitative output and neural-derivative allowances, and a structured continuous-action oracle. The unchanged coupled comparison problem has an absolute lower reference and a complete frozen holdout record, including unfavorable baseline and sign-test outcomes. Those experiments do not establish a universally best-tuned neural solver or an external matched-accuracy dominance theorem. The added reference experiments give certified continuous-model accuracy, implemented-output error identities, and computation-to-accuracy paths rather than only fixed-budget losses. The expanded tuning study assesses robustness under a second frozen protocol, not by retuning the original holdout. These additional results strengthen the executable theory–algorithm link while leaving each certificate attached to its actual model, policy class, and implementation. All preceding manuscripts, equations, negative diagnostics, and raw outcomes remain preserved for further examination.

APPENDIX A: PROOFS FOR CONTINUOUS-COST CERTIFICATION

A.1. Proof of Theorem 6

Fix a policy π as a progressively measurable rule, including its response to the state and its stopping convention. Changing only k does not change its state law or liquidation time. Linearity of expectation therefore gives

$$J_k(\pi) = J_{k_j}(\pi) - (k - k_j)B(\pi).$$

The integrals are finite: the state and action sets before stopping are bounded, the consumption lower bound is positive, risk aversion stays bounded away from one, and the liquidation payoff is bounded on the stopped trace. For $k \geq k_j$, replace J_{k_j} by its lower bound and B by its upper bound; for $k < k_j$, replace B by its lower bound. This proves $\ell_j(k) \leq J_k(\pi_j)$ on both sides, including equality of the formulas at the node.

The same common-policy-set argument writes $V(k) = \sup_{\pi} \{A(\pi) - kB(\pi)\}$. A supremum of affine functions is convex, whether or not a maximizer exists. Hence the chord between $V(k_i)$ and $V(k_{i+1})$ bounds V from above; replacing both endpoints by valid uppers proves $V(k) \leq u(k)$. The library is finite, so a lowest-index maximizer exists. Combining its feasible-policy lower with this chord proves (39).

Inside a node interval no k_j changes the sign of $k - k_j$. Each ℓ_j is therefore affine there. Their upper envelope is continuous, convex, and piecewise affine. Subtracting it from the affine chord gives a continuous piecewise-affine function, whose maximum on each linear piece occurs at an endpoint (or everywhere on a constant piece). Its candidates are thus the cell endpoints and envelope breakpoints. For rational input coefficients, every intersection of unequal slopes is rational. Equal-slope lines are reduced to the highest intercept, with a deterministic tie rule. Sorting by slope and successively removing lines whose intersection does not increase constructs the upper envelope. The acceptance calculation uses exact rational arithmetic throughout; a binary64 endpoint is treated as the exact rational number it represents, not as a decimal approximation to a new endpoint. The final decimal display is rounded upward. $\square$

A.2. Proof of Proposition 3

Write $a = k_i$, $b = k_{i+1}$, $h = b - a$, and $s = (k - a)/h$. As $\ell \geq \max\{\ell_i, \ell_{i+1}\}$, it also exceeds their convex combination with weights $1 - s, s$. The expenditure endpoints enter as

$$\ell_i(k) = L_i - shb_i^+, \quad \ell_{i+1}(k) = L_{i+1} + (1 - s)hb_{i+1}^-.$$

Consequently,

$$\begin{aligned} u(k) - \ell(k) &\leq (1 - s)(U_i - L_i) + s(U_{i+1} - L_{i+1}) + s(1 - s)h(b_i^+ - b_{i+1}^-) \\ &\leq \max\{U_i - L_i, U_{i+1} - L_{i+1}\} + \bar{B}h/4. \end{aligned}$$

This proves (41). A finite number of dyadic subdivisions makes all lengths strictly less than $4\eta/\bar{B}$. Fair processing of failed intervals then terminates, conditional on the assumed local slack. No conclusion about the training algorithm's rate or ability to attain that slack follows from this argument. $\square$

A.3. Cost-transfer and stopping details

For the deployed policies $p = 0$, consumption is budget-projected into the original action interval and preference drift lies in $[0, .2]$. Wealth does not cross the lower liquidation face before the horizon, as verified independently by the inherited budget certificate. Preference exit is not discarded. The moment evaluator subtracts an explicit stopped-payoff error and encloses the actual discounted expenditure in

$$[B_T - .02 \Pr\{\tau < T\}, B_T], \quad B_T = \int_0^1 e^{-.04t} \theta_t^2 / 2 dt,$$

with $\Pr\{\tau < T\} \leq 2e^{-72}$. These outward endpoints, rather than the deterministic full-horizon expenditure alone, supply $b_j^\pm$ in the continuation rule. The stored controls are binary64 real constants strictly within the exact action constraints. No new Euler or Brownian-bridge error enters their continuous stopped evaluation.

For (42), subtract the two valid value intervals and intersect with monotonicity and the primitive expenditure bound. For (43), use $V(q) \geq J_q(\pi_k^*) = V(k) - (q - k)B(\pi_k^*)$ and the appropriate value endpoints. The upper bound follows in the same way from $p < k$. Every intersection is between valid bounds on the same object; the existence qualification concerns the optimal policy's budget only, not the value or policy-regret guarantee.

A.4. Complete refinement and interruption record

The pilot at $k = 4$ follows the three inherited certificates. Subsequent completed node sets have sizes seven, thirteen, seventeen, and eighteen. Two multi-batch local invocations were interrupted by the execution tool before their last cells completed. Their partial files and status receipts remain in the revision archive. The missing prices were explicitly retried and individually verified; no incomplete case was accepted. The final envelope requires every planned price to have a completed certificate. Failure preservation and acceptance are distinct: retaining a failed attempt does not invalidate a subsequently certified fixed policy, while a missing final certificate prevents a complete-package pass.

The accompanying code tests sign-sensitive transfer on both sides of a node, exact hull values against direct evaluation, breakpoint maxima against an independent all-pairs construction, deliberate missing and nonfinite cases, and the conditional refinement inequality. Such tests detect implementation mistakes; the proofs above, together with the original interval evaluators, establish the mathematical enclosure. Machine-dependent timing is never an acceptance criterion.

APPENDIX B: CONTINUOUS REFERENCES, HELD-ACTION MOMENTS, AND OUTPUT-ERROR IDENTITIES

This appendix proves Proposition 5 and specifies the complete arithmetic used in Section 10. It is independent of the stopped-economy dual. All quantities here concern the linear-quadratic objective (50); no claim about a constrained nonquadratic optimum is obtained by replacing its payoff with this one.

B.1. Continuous verification and the closed reference

We give a scalar proof; summation and division by d yield the stated result. The finite-horizon Riccati equation with $q \geq 0$, $r, w > 0$ has a nonnegative bounded solution. This can also

TABLE A.I
ALL COST-CONTINUATION NODE CERTIFICATES

k_j	Feasible value lower	Optimal value upper	b_j^-	b_j^+
0.5	-1.27267126	-1.26279686	0.01757696	0.01757697
0.875	-1.27898110	-1.26923395	0.01615106	0.01615107
1.25	-1.28480062	-1.27508670	0.01486625	0.01486626
1.625	-1.29014098	-1.28043412	0.01362631	0.01362632
2	-1.29502451	-1.28530723	0.01241866	0.01241867
2.5	-1.30084578	-1.29110538	0.01088620	0.01088621
3	-1.30591113	-1.29614160	0.00940418	0.00940419
3.5	-1.31025400	-1.30045120	0.00797493	0.00797494
4	-1.31390273	-1.30406565	0.00662588	0.00662589
4.25	-1.31547447	-1.30562313	0.00596692	0.00596693
4.5	-1.31689371	-1.30702987	0.00540238	0.00540239
5	-1.31935619	-1.30947023	0.00449104	0.00449105
5.5	-1.32141969	-1.31151486	0.00379351	0.00379352
6	-1.32317442	-1.31325326	0.00324749	0.00324750
6.5	-1.32468517	-1.31474973	0.00281191	0.00281192
7	-1.32599973	-1.31605175	0.00245877	0.00245878
7.5	-1.32715413	-1.31719506	0.00216844	0.00216845
8	-1.32817604	-1.31820708	0.00192681	0.00192682

All displayed endpoints are rounded outward. The expenditure is that of the specified feasible policy under actual stopping. Full-precision endpoints, policy controls, dual coefficients, and arithmetic allowances are retained in the per-price records. Three nodes are inherited and fifteen are new.

be checked directly from the displayed reference formula: it is the fractional transformation generated by

$$\frac{d}{ds} \begin{pmatrix} A(s) \\ B(s) \end{pmatrix} = \begin{pmatrix} -a & c \\ q & a \end{pmatrix} \begin{pmatrix} A(s) \\ B(s) \end{pmatrix}, \quad (A(0), B(0)) = (1, w), \quad c = b^2/r.$$

The matrix square is $(a^2 + cq)I$, giving the hyperbolic formula in the text. With $P(s) = B(s)/A(s)$, differentiation gives $P' = q + 2aP - cP^2$. Nonnegative initial data stay nonnegative, and the negative quadratic term prevents a finite-time upward explosion. Moreover $A' = (-a + cP)A$ and $A(0) = 1$, so A stays positive and

$$\int_0^1 P(s)ds = \frac{\log A(1) + a}{c}.$$

At $\lambda = 0$ in the specified cases, $a = q = 0$, $A(s) = 1 + cws$, and $B(s) = w$. This branch is evaluated by its exact limit, not by division by a small interval containing zero.

Set $v(t, x) = p(t)x^2 + \sigma^2 \int_t^1 p(s)ds$, where $p(t) = P(1 - t)$. Direct substitution gives

$$v_t + (ax + bu)v_x + \frac{1}{2}\sigma^2 v_{xx} + qx^2 + ru^2 = r \left(u + \frac{bp(t)}{r}x \right)^2.$$

For a square-integrable adapted control on a finite horizon, the linear state equation has finite integrated and terminal second moments. Localizing the Itô integral and passing to the limit in the nonnegative quadratic terms gives the verification identity (52); equivalently one can first prove it for bounded stopped controls and use the explicit linear state solution and L^2 approximation. The feedback $u = -bp(t)X_t/r$ is admissible, has finite second moments, and makes the square vanish. It therefore attains the value in (51). This proves both directions of optimality rather than using a feasible Riccati policy as a lower bound without justification.

B.2. Exact deployment coefficients

For one interval of length h , put

$$E = e^{ah}, \quad I = \int_0^h e^{as} ds, \quad A_2 = \int_0^h e^{2as} ds, \quad B_2 = \int_0^h e^{as} \left(\int_0^s e^{av} dv \right) ds,$$

$$C_2 = \int_0^h \left(\int_0^s e^{av} dv \right)^2 ds, \quad D_2 = \int_0^h \int_0^s e^{2av} dv ds.$$

For $a \neq 0$, these quantities are

$$I = (e^{ah} - 1)/a, \quad A_2 = (e^{2ah} - 1)/(2a),$$

$$B_2 = (A_2 - I)/a, \quad C_2 = (A_2 - 2I + h)/a^2, \quad D_2 = (A_2 - h)/(2a).$$

For $a = 0$, they are $I = A_2 = h$, $B_2 = D_2 = h^2/2$, and $C_2 = h^3/3$. Sixty-decimal interval evaluation, with a separate zero-drift branch, encloses cancellation in these expressions. All actual nonzero drift inputs in the finite suite are point intervals bounded away from zero; a division by an interval containing zero is rejected rather than repaired numerically.

Conditional on $X_{ih} = x$ and an action u held through the interval,

$$\mathbb{E}[X_{(i+1)h}^2 \mid x, u] = (Ex + bIu)^2 + \sigma^2 A_2,$$

$$\mathbb{E} \left[\int_{ih}^{(i+1)h} (qX_t^2 + ru^2) dt \mid x, u \right] = qA_2 x^2 + 2qbB_2 xu + (qb^2 C_2 + rh)u^2 + q\sigma^2 D_2.$$

The formulas retain the correlation between the held action and the state at the beginning of the slab. They do not approximate the action by a continuous feedback evaluated along the evolving state.

If the remaining value after this slab is $P_{i+1}x^2 + z_{i+1}$, define

$$D_i = qb^2 C_2 + rh + P_{i+1}b^2 I^2 > 0,$$

$$L_i = qbB_2 + P_{i+1}EbI, \quad K_i^* = L_i/D_i,$$

$$P_i = qA_2 + P_{i+1}E^2 - L_i^2/D_i,$$

$$z_i = z_{i+1} + q\sigma^2 D_2 + P_{i+1}\sigma^2 A_2, \quad P_N = w, \quad z_N = 0.$$

The Bellman minimization is exact completion of the square. Nonnegativity of costs implies $P_i \geq 0$, so the strict inequality for D_i follows from $rh > 0$. It applies to the entire unbounded action line, not merely to a stationary point. These N backward steps are closed exact evaluation and improvement operations for quadratic values and linear held-action outputs.

For any adapted held-action rule, its conditional Bellman excess at stage i is $D_i(u_i + K_i^* X_{ih})^2$. Telescoping the conditional expectations gives

$$J(u) - J_h^* = \frac{1}{d} \sum_{i,j} D_{ij} \mathbb{E}[(u_i^j + K_{ij}^* X_{ih}^j)^2].$$

Combining this equality with the continuous verification result proves (53). In particular, for an implemented linear actor $u_i^j = -K_{ij} X_{ih}^j$, the optimization error is weighted by the *deployed*

actor's second moments, not by moments under an ideal policy. If $|K_{ij} - K_{ij}^*| \leq \delta_{ij}$ and these moments are at most M_{ij} , the explicit allowance is $d^{-1} \sum D_{ij} \delta_{ij}^2 M_{ij}$. This is an output-level certificate for the implemented optimizer; it is not a statement that its weight-space objective is convex.

B.3. Independent interval policy evaluation

The verifier does not reuse the backward value as the value of a stored floating-point policy. Instead, starting with $m_0 = x_0^2$, it propagates

$$\begin{aligned} m_{i+1} &= (E - bIK_i)^2 m_i + \sigma^2 A_2, \\ \Delta J_i &= \{q(A_2 - 2bK_i B_2 + b^2 K_i^2 C_2) + rhK_i^2\} m_i + q\sigma^2 D_2, \\ J(K) &= \sum_i \Delta J_i + w m_N. \end{aligned}$$

Each coefficient and stored gain is imported as the exact binary64 value that defines the deployed model and actor. The arithmetic is outward at 60 decimal digits. Consequently the policy-cost interval and the closed-reference interval enclose two independently computed original-model quantities. Subtraction in interval arithmetic yields an absolute-regret interval; a negative upper endpoint is an error and is not clipped to zero. Conversion to machine-readable binary64 endpoints expands the lower endpoint toward $-\infty$ and the upper endpoint toward $+\infty$ by one unit. The output files hash the little-endian binary64 gain array, binding every certificate to its policy.

The differentiable Adam objective is a separate torch implementation of the moment equations in binary64. Its agreement with the interval evaluator is checked to a small conversion tolerance, but that agreement is only a software cross-check. The interval equations and continuous-reference proof, not the tolerance in that test, establish the certificate. Additional tests check the closed formula against a separately integrated Riccati equation, test the scalar tracking formula, reject nonfinite gains, and verify that a deliberately perturbed actor changes both its policy hash and cost. Timings and arithmetic intervals are kept separate: timing fluctuations cannot change whether a stored policy is certified at the prescribed tolerance.

APPENDIX C: PROOF OF THE AFFINE-PREFERENCE DUAL

This appendix gives the analytic inequalities behind the new upper bound and identifies exactly which finite interval calculations enter the proof. The frozen fitting objectives are not proof inputs. The only fitted objects subsequently used are the exact stored values of y_0 and the sixteen reference drifts, whose admissible ranges are checked independently.

C.1. The supporting-plane inequality

Let $r = u - 1$, $l = \log y$, and

$$L = \text{proj}_{[-\log(.8), -\log(.05)]}(l/u), \quad c = e^{-L}, \quad E = e^{Lr}.$$

The consumption maximizer is c , including both constrained branches. Write $\chi = 1$ on the interior branch and zero on a constrained branch; on a box intersecting a switching surface its

interval enclosure is $[0, 1]$. The derivatives required by the certifier are

$$\begin{aligned}
\Psi &= -E/r - yc, & \beta &= E(1 - Lr)/r^2, \\
(S_0)_l &= -yc + .01y, \\
(S_0)_{uu} &= -\frac{E\{(Lr - 1)^2 + 1\}}{r^3} + \chi \frac{ycL^2}{u}, \\
(S_0)_{ul} &= -\chi \frac{ycL}{u}, \\
(S_0)_u &= -yc + \chi \frac{yc}{u} + .01y.
\end{aligned} \tag{54}$$

On the interior branch $yc = E$. The envelope is continuously differentiable across the consumption constraint; the displayed one-sided second derivatives enclose its almost-everywhere Hessian. This is sufficient both for supporting inequalities and for the integral Taylor remainder used below.

No lower-consumption constraint is active on $u \in [1.5, 2.5]$, $y \in [.2, 12]$, because $\log(12)/1.5 < -\log(.05)$. Moreover $L < 2$ throughout that rectangle. On its interior consumption branch,

$$\partial_l(S_0)_{uu} = \partial_u \beta_l = \frac{EL(2 - L)}{u^2} \geq 0. \tag{55}$$

For the upper-consumption branch $(S_0)_{uu}$ is independent of l and $\beta_l = 0$. The jump in $(S_0)_{uu}$ when increasing l enters the interior branch is nonnegative. When increasing u leaves that branch, β_l jumps upward to zero. Thus $(S_0)_{uu}$ is nondecreasing in l , and β_l is nondecreasing in u , including the switching boundary.

The following enclosures are computed by the interval program on complete interval covers, not by point sampling:

$$\sup_{1.5 \leq u \leq 2.3} (S_0)_{uu}(u, 12) < -.0862, \tag{56}$$

$$\sup_{2.3 \leq u \leq 2.5} \beta(u, 12) < -.9687, \tag{57}$$

$$\beta(2.2, 12) > -.9573. \tag{58}$$

The first cover has 1,600 intervals and the second has 2,000. All decimal endpoints and analytic constants are enclosed at 70-digit interval precision; all subsequent binary operations and interval hulls are rounded outward. Full endpoints and the finite cover prescription are stored in `tangent_primitive_checks.json` and its generating program.

By (55), the first inequality proves concavity on $[1.5, 2.3]$ for every y in the verification rectangle. For $u \in [2.3, 2.5]$ and $m \in [2, 2.2]$, the difference $\beta(u, y) - \beta(m, y)$ is nondecreasing in l , since β_l is nondecreasing in u . Its maximum therefore occurs at $y = 12$. At that endpoint, concavity gives $\beta(m, 12) \geq \beta(2.2, 12)$, and the last two inequalities in (58) make the difference strictly negative. Integrating the derivative between m and u now proves (15) on both sides of m . This argument does not infer a two-dimensional supporting plane from concavity at a single endpoint without controlling the cross derivative.

The same program covers $m \in [2, 2.2]$ by 1,000 intervals and uses $(S_0)_l < 0$, $\beta_l \leq 0$ to check the two y endpoints. It obtains

$$-6.989 < S_0(m, y) < -1.2443, \quad |\beta(m, y)| < .9712. \tag{59}$$

The deliberately simpler bounds $-7.1 \leq S_0 \leq 0$ and $|\beta| \leq 1.1$ are used for tails and localization. They continue to hold after clipping y .

C.2. An adapted solution of the affine auxiliary problem

For this subsection start at $(0, 2, y_0)$ and use the full-line auxiliary processes. All sources are affine in U , with bounded coefficients, and all controls are bounded. Fubini's theorem, conditional expectation, and optional stopping on bounded rectangles are legitimate by Gaussian first-moment bounds. With

$$B_s = \int_s^1 e^{-\rho(t-s)} \mathbb{E}[\beta_t | \mathcal{F}_s] dt + e^{-\rho(1-s)} b_T,$$

Markov independence of future Brownian increments gives $\mathbb{E}[\beta_t | \mathcal{F}_s] = \mathbb{E}[\beta_t | Y_s]$, even when the control observes the full correlated Brownian history. In particular, B_s is an adapted function of (s, Y_s) and $\mathbb{E}B_s = b(s)$.

Write

$$U_t - m(t) = \int_0^t (\theta_s - \bar{\theta}_s) ds + .05 Z_t^u.$$

Substitution in (16), followed by conditional Fubini, shows that its control-dependent part is

$$\mathbb{E} \int_0^1 e^{-\rho s} \{(\theta_s - \bar{\theta}_s) B_s - k\theta_s^2/2\} ds.$$

The pointwise maximizer $\theta_s = \text{proj}_{[-.2, .2]}(B_s/k)$ is adapted and bounded. It gives a well-defined auxiliary U because it is a function of the exogenous process Y . Thus this maximization introduces neither perfect foresight nor a new assumption about the optimizer of the original economy. Its exact value yields

$$\begin{aligned} A_k(0, 2, y_0) &= e^{-\rho} \phi(m(1)) + \int_0^1 e^{-\rho t} \{\mathbb{E}S_0(m(t), \hat{Y}_t) - k\bar{\theta}_t^2/2\} dt + C \\ &\quad + \int_0^1 e^{-\rho s} \mathbb{E}g_k(B_s, \bar{\theta}_s) ds. \end{aligned} \tag{60}$$

For the stochastic source term, the joint Gaussian covariance is

$$\text{Cov}(.05Z_t^u, \log Y_t) = (.05)(-.3)(-.25)t = .00375t.$$

Gaussian integration by parts consequently gives exactly the term C in (22). The clipping and consumption switches are continuous Lipschitz maps; their weak derivatives, rather than nonexistent classical derivatives at isolated switches, suffice for this identity. Since $\beta_t \leq 0$, $C \leq 0$. In particular, treating the two noises as independent would discard a real term of the original economy.

C.3. The conditional-variance allowance

The function H_k is continuously differentiable, with $H'_k(q) = \text{proj}_{[-.2, .2]}(q/k)$. Its derivative is $1/k$ -Lipschitz. The same bound applies to the derivative of $q \mapsto g_k(q, a)$ for fixed a . Therefore

$$\mathbb{E}g_k(B_s, \bar{\theta}_s) \leq g_k(b(s), \bar{\theta}_s) + \frac{\text{Var}(B_s)}{2k}. \tag{61}$$

Let $l_s = \log Y_s$ and write $B_s = f_s(l_s)$. The law of l_s is Gaussian with mean $M - .025s$ and variance $.09s$. The Gaussian Poincare inequality gives

$$\text{Var}(B_s) \leq .09s \mathbb{E}|f'_s(l_s)|^2.$$

For completeness, with an orthonormal Hermite expansion $f = \sum_{n \geq 0} c_n H_n$ under a Gaussian of variance v , one has $\text{Var} f = \sum_{n \geq 1} c_n^2$ and $v \mathbb{E}|f'|^2 = \sum_{n \geq 1} n c_n^2$. Polynomial approximation extends the inequality to the bounded Lipschitz functions used here. No unknown mixing or concentration constant enters the calculation.

Differentiation of the conditional Gaussian transition is a translation derivative, so

$$f'_s(l_s) = \int_s^1 e^{-\rho(t-s)} \mathbb{E}[\partial_t \beta_t | Y_s] dt.$$

Conditional Jensen and Minkowski give

$$\text{Var}(B_s) \leq .09s \left\{ \int_s^1 e^{-\rho(t-s)} \sqrt{\mathbb{E}|\partial_t \beta_t|^2} dt \right\}^2 \leq .09s D^2 C_\rho (1-s)^2. \quad (62)$$

We next verify the constant in this inequality. On the interior branch and before clipping, for $m \in [2, 2.2]$,

$$|\partial_l \beta(m, e^l)|^2 = e^{2(1-1/m)l} \frac{l^2}{m^4} \leq \frac{1}{16} e^{(12/11)l} l^2 \quad (l > 0).$$

For $l \leq 0$, the upper-consumption constraint makes the derivative zero. At a clipped y face the derivative is also zero almost everywhere. The positive right-hand side can therefore be integrated over the entire untruncated Gaussian as an upper bound. Its exact exponential moment is

$$\mathbb{E}[e^{al^2}] = e^{a\mu + a^2 v/2} \{(\mu + av)^2 + v\}.$$

For $y_0 \in [1.2, 2.4]$, the means $\mu = M - .025t$ are positive. The expression is increasing in both μ and v on the relevant range. Using $\mu \leq M$ and $v \leq .09$ gives (24). Finally,

$$\int_0^1 e^{-\rho s} s C_\rho (1-s)^2 ds \leq \int_0^1 s(1-s)^2 ds = 1/12.$$

Combining (60), (61), and (62) proves (23). The allowance is positive and remains present at every quadrature resolution.

C.4. Transfer from the auxiliary problem to the actual contract

Set

$$W(t, u, x, y) = y(x - .5) + .1 \log(.5) + A_k(t, u, y).$$

The accounting identity for Yx cancels every original adaptive portfolio choice. The maximization defining Ψ bounds every original consumption. Inequality (15) bounds the remaining source as long as (u, y) lies in the verification rectangle. Dynamic programming for the explicitly solved affine auxiliary problem then makes discounted W plus accumulated original payoff a supermartingale up to the first relevant face, for any original policy. One can establish the same statement directly from conditional Fubini and the adapted selector above; no differentiability of

the unknown original value is needed. Smooth approximations of the bounded Lipschitz source, with uniform source-error corrections, give the equivalent classical verification argument.

The terminal tangent obeys $Q \geq \phi$ everywhere because ϕ is concave. Write $F = A_k - Q$. Under the feasible auxiliary choice $\theta = 0$, discounted Ito applied to the affine function Q gives

$$F(t, u, y) \geq \mathbb{E} \int_t^1 e^{-\rho(r-t)} \{S_0(m(r), \hat{Y}_r) + \beta(m(r), \hat{Y}_r)(U_r - m(r)) - \rho Q(U_r)\} dr.$$

For any starting $u \in [1.5, 2.5]$, one has

$$\mathbb{E}|U_r - m(r)| \leq .7 + .05 = .75, \quad \mathbb{E}|Q(U_r)| \leq .0008 + .008(.75) = .0068.$$

Consequently (59) implies

$$F(t, u, y) \geq \{-7.1 - 1.1(.75) - .04(.0068)\}(1-t) = -7.925272(1-t) > -8(1-t). \quad (63)$$

This is a uniform bound for the required starting rectangle; it is not a claim that the affine potential is bounded on the entire u line.

For $x \in [.5, 2]$ and $y \geq .2$,

$$y(x - .5) + .1 \log(.5) - .1 \log x \geq 0,$$

by differentiation in x , with equality at $x = .5$. Thus $W \geq G - 8h$ on actual wealth exits and $W \geq G$ at $h = 0$. For the preference localization, $.2(120) + .00125(120)^2 = 42$ bounds the killed generator for either sign and every original θ . For log-deflator localization, $dl = -.025dt - .3dZ^x$ and $22.33 = .025(22) + .045(22)^2$ bounds either sign. Each term in (18) is a nonnegative supermartingale potential that contributes at least $8h$ on its artificial face. Their addition therefore raises the continuation value to at least G at every artificial face. The already established global upper $V_k \leq G$, for $k \geq .5$, applies beyond that face. Stopped verification before it and this fallback after it prove (19). All constants correspond to the original discount, diffusion correlation, action bounds, and liquidation fee.

APPENDIX D: VALIDATED EVALUATION AND IMPLEMENTATION ERROR

D.1. Interval quadrature of the dual

The frozen references have sixteen time slabs. A slab $t \in [j/16, (j+1)/16]$ is transformed to $v \in [\sqrt{j/16}, \sqrt{(j+1)/16}]$ and subdivided into 4, 8, or 16 intervals. The standard Gaussian is covered by 64, 128, or 256 intervals on $[-6, 6]$, respectively. Thus the three meshes use 4,096, 16,384, and 65,536 source boxes. All square-root endpoints are enclosed before binary64 conversion.

On a box the reference mean and log-deflator are

$$m(v^2) = 2 + \mu_j + \bar{\theta}_j(v^2 - j/16), \quad l(v, z) = M - .025v^2 - .3vz,$$

where $\mu_j = \sum_{i < j} \bar{\theta}_i/16$ is an interval-enclosed sum of exact deployment constants. The source Hessian in (v, z) follows from (54) and the chain rule. The entire image box is enclosed, with the hull of all possible consumption branches. The program asserts that the covered Gaussian region stays inside the y clipping faces for each of the fitted references.

With $w_v = 2ve^{-.04v^2}$ and $w_z = (2\pi)^{-1/2}e^{-z^2/2}$, the source integral uses the exact interval-enclosed zeroth, first-centered, and second-centered moments of both weights. The exponential

series for the time weight retains terms through degree twelve in $-.04v^2$ and bounds the remaining terms analytically. The Gaussian weight uses its degree-110 exponential series with an analytic absolute remainder on $[-6, 6]$. Antiderivatives of each polynomial term are evaluated at interval endpoints. No floating-point Gaussian quadrature weight is assumed exact in the certificate.

If r_v, r_z are box radii, a valid second-order remainder bound is

$$\frac{1}{2}\|S_{vv}\|_\infty M_{v,2}M_{z,0} + \|S_{vz}\|_\infty r_v r_z M_{v,0}M_{z,0} + \frac{1}{2}\|S_{zz}\|_\infty M_{v,0}M_{z,2}.$$

The interval first-centered moments retain their signs; constant and linear terms are not discarded as though all weighted boxes were symmetric. The source is C^1 with locally bounded one-sided Hessians across an active-consumption switch, so the integral Taylor formula with the Hessian hull remains valid there.

A Gaussian Chernoff bound gives $\Pr(|Z| > 6) \leq 2e^{-18}$. The source tail lies between $-7.1\Pr(|Z| > 6)$ and zero at each time; the coefficient tail has magnitude at most $1.1\Pr(|Z| > 6)$. For the covariance tail,

$$|\partial_t \beta| \leq 12^{6/11} \log(12)/4 < 2.5$$

on the nonzero-derivative region, so its omitted contribution lies in $[-.00375(2.5)\Pr(|Z| > 6)/2, 0]$. The signs are retained because $\beta_t \leq 0$. These corrections account for the full unbounded Gaussian, not just the sampled central region.

For $b(s)$, interval source-coefficient integrals are accumulated backwards. The remaining fraction of the current time cell ranges independently from zero to its entire discounted mass. Multiplying by the interval for $e^{\rho s}$ encloses $b(s)$ throughout that cell. The convexity of $g_k(q, \bar{\theta})$ makes the maximum over its resulting q interval occur at an endpoint. At an uncertain saturation threshold the unconstrained quadratic conjugate supplies a safe upper bound. This calculation supplies the control-gap integral in (23).

Every elementary binary64 addition, subtraction, multiplication, division, and pairwise sum is rounded outward. Analytic constants and weighted moments use 70-digit interval arithmetic. The fitting objective, the source-quadrature enclosure, the policy-payoff enclosure, and the optimality gap have different meanings and are recorded in different files. The acceptance test subtracts the policy's lower endpoint from the optimal-value upper endpoint with outward rounding. The comparison with .01 is then a test of the original-policy regret, not of a residual norm.

D.2. Proof of the feasible-output allowance

PROOF OF PROPOSITION 2: Couple the two policies to the same preference Brownian motion. Wealth is deterministic, decreasing, and remains in $(.5, 1.25]$ because of the stated budget and consumption constraints. The difference of preference paths at time t is at most $\int_0^t |\theta_s - \tilde{\theta}_s| ds$. The terminal wealth difference is at most $e^{.02} \delta_c$.

On the event that neither policy exits through preference before time one, $u \in [1.2, 2.8]$. For $c \in [.7, .8]$, put $r = u - 1$ and $L = -\log c$. Since $Lr < 1$, the running utility satisfies

$$|\partial_c \ell| = c^{-u} < 3, \quad 0 \leq \partial_u \{c^{1-u}/(1-u)\} = e^{Lr}(1-Lr)/r^2 \leq 1/r^2 \leq 25.$$

The terminal preference derivative has magnitude at most .032, while the terminal wealth derivative is at most .2. Finally, $\frac{k}{2}|\theta^2 - \tilde{\theta}^2| \leq .2k|\theta - \tilde{\theta}|$. Integrating these inequalities and using discount factors at most one gives the first two terms of (25).

Each policy's preference exit has probability at most $2e^{-72}$ by the same maximal inequality used in the independent evaluator. The union thus has probability at most $4e^{-72}$. On any stopped path, utility has magnitude below 6 before the adjustment cost, that cost contributes at most $.02k$, and settlement has magnitude below 8.1. Each payoff therefore has absolute value at most $14.1 + .02k$. Bounding the payoff difference on the exit union by twice this quantity gives the last term of (25). The argument never replaces the stopped economy with a reflected or unstopped payoff. *Q.E.D.*

D.3. What the finite execution establishes

All three cost instances pass the .01 acceptance test. The interval upper bounds apply to arbitrary original adaptive controls; the lower bounds apply to the stored polished time policies. This distinction is essential to an optimality certificate. The computations do not identify a globally optimal feedback surface, establish a rate for the neural optimizer, or prove that a fixed affine relaxation converges to exact value as its quadrature mesh tends to zero. Its nonvanishing allowances are displayed explicitly. The finite target and the economic inequalities reported in the main text follow from the executed upper and lower bounds alone.

APPENDIX E: THE PRECEDING FITTED CONSTRUCTION, RETAINED FOR COMPARISON

The following section is preserved verbatim from the completed R9 paper. Its policies, upper witnesses, tables, and statements that the .01 target was not met describe that preceding computation. The current R10 certificate is Theorem 3 and Table I. This dated preservation is not an alternative current claim.

APPENDIX F: CONSTRUCTIVE CERTIFICATES IN THE ORIGINAL ECONOMY

All initial numerical bounds in this section concern $(t, u, x) = (0, 2, 1.25)$. A central-state certificate is not advertised as a uniform certificate for every state in D . The policy used here is newly trained in a time-control neural subclass. The retained lattice-trained, state-feedback NBO policies, their finite corrections, and their previous continuous bounds remain separately available in the supplement.

F.1. Fitting an upper witness without importing an optimal value

Put $h = 1 - t$ and restrict the *verification domain*, not the economic domain, to $u \in [1.5, 2.6]$.
Fit

$$F(h, u) = h \sum_{i=0}^{12} \sum_{j=0}^{14} \beta_{ij} T_i(2h - 1) T_j \left(2 \frac{u - 1.5}{1.1} - 1 \right).$$

The coefficients are obtained from a method-of-lines initializer and residual least squares. The optimizer's collocation residual is not the certificate. For $w = G + F$, the consumption and portfolio maximizers are .8 and .8. Maximizing the remaining wealth term over $x \in [.5, 2]$ gives the scalar residual majorant

$$\mathcal{R}_k F = -F_h + .00125 F_{uu} - .04 F + H_k \{F_u - .04(u - 2)\} + \frac{.8^{1-u}}{1-u} \quad (64)$$

$$+ .0008(u - 2)^2 + .00195 + .00352 - .04 - .004 \log 2, \quad (65)$$

$$H_k(q) = \max_{|\theta| \leq .2} (q\theta - k\theta^2/2).$$

The source is the original utility, not a payoff manufactured to make F exact.

Power coefficients are enclosed at 70-digit interval precision and converted to Bernstein coefficients. The utility term is replaced by a polynomial with a proved uniform remainder below 2.117×10^{-9} . De Casteljau subdivision uses outward rounding at every elementary step. The positive-saturation, negative-saturation, and interior branches of H_k are all included whenever their certified q range permits them. Outward thresholds also cover a cell touching a switching surface. The same calculation proves $F/h \geq -8$.

PROPOSITION 6—Localized fitted upper witness: *Suppose $F(0, u) = 0$, $F/h \geq -8$, and $\mathcal{R}_k F \leq \epsilon$ on the verification rectangle. For $k \geq .5$, let*

$$Z(h, u) = 8h\{e^{-120(u-1.5)+42h} + e^{-160(2.6-u)+64h}\}, \quad (66)$$

$$U(t, u, x) = G(u, x) + F(1-t, u) + Z(1-t, u) + \epsilon_+ C_\rho(1-t). \quad (67)$$

Then $V_k(0, 2, 1.25) \leq U(0, 2, 1.25)$. No reflecting boundary or truncation of the original policy is introduced at $u = 1.5$ or 2.6 .

The exponential terms bound continuation when an artificial verification face is reached. The globally valid upper witness G is used beyond that face. At the central state the localization addition is less than 1.219×10^{-7} , rather than an eight-unit initial gap. Uniform residual correction and localization are different terms and are reported separately.

F.2. A learned feasible policy and independent stopped evaluation

The actor has one time input, a twelve-unit tanh layer, and two outputs. At sixteen fixed time nodes, a sigmoid constrains adjustment to $[0, .2]$, and a differentiable budget projection constrains consumption to $[.05, .8]$. The portfolio is zero. Training uses the original expected payoff, Gaussian quadrature, and no optimal-value or greedy-action labels. The deployed controls are the resulting frozen binary64 constants on each time slab; there is no claim that this time-control class exhausts the state-feedback opportunities in (5).

The budget projection enforces

$$\sum_{j=0}^{15} c_j \int_{j/16}^{(j+1)/16} e^{-.02r} dr < 1.25 - .5e^{-.02}. \quad (68)$$

Thus $x_t = e^{.02t} \{1.25 - \int_0^t e^{-.02r} c_r dr\}$ is deterministic. Since consumption exceeds $.02(1.25)$, wealth decreases and its interval-enclosed terminal value remains strictly above $.5$. Wealth never exits. The remaining preference exit is controlled by a Brownian maximal inequality, not by a discretized boundary check.

An evaluator independent of the training quadrature approximates utility by a degree-90 polynomial in $u - 2$, with a uniform analytic remainder on the original preference interval. Free Gaussian moments are evaluated by recurrence, integrated over outward time rectangles, and corrected for the stopped event using a maximal probability bound and Cauchy–Schwarz. The terminal wealth, adjustment expenditure, polynomial approximation, time integration, exact-exit correction, and arithmetic are all enclosed. There is no Euler, action-grid, or spatial-interpolation term in this route. Those omissions are consequences of the deployed policy and analytic evaluation, not uncomputed entries assigned zero.

Table F.I reports the actual original-economy bounds. Table F.II makes the constructive refinement explicit. The last row gives substantially more precise policy evaluation than optimality

TABLE F.I
ORIGINAL-PAYOFF CONTINUOUS CERTIFICATES AT THE CENTRAL STATE

k	Policy payoff enclosure	Optimal-value upper	Regret upper
0.5	$[-1.27324522, -1.27322724]$	-1.16618072	0.10706450
2	$[-1.29504316, -1.29502986]$	-1.18870455	0.10633861
8	$[-1.32818098, -1.32817412]$	-1.22254633	0.10563464

Payoff endpoints and upper bounds are rounded outward. Each regret upper is the independent optimal-value upper minus the lower endpoint of the learned policy's payoff interval. All controls in the optimal-value bound are continuous. The .01 full-policy target is not met in any row.

TABLE F.II
EXECUTED REFINEMENT FOR THE $k = 2$ CERTIFICATE

Time boxes	Payoff width	Sec.	Bernstein boxes	Residual upper	Sec.	Regret upper
1024	0.000209	0.363	36	0.00477	0.079	0.110237
4096	5.25e-05	0.614	57	0.00195	0.116	0.107396
16384	1.33e-05	1.618	86	0.000894	0.186	0.106339

Time boxes refine the independent policy evaluator. Bernstein boxes refine the already fitted upper witness. The two seconds columns measure the respective integration and subdivision stages; complete timings, including coefficient construction, are in the raw records. A shrinking evaluation or residual allowance need not eliminate the approximation gap between the chosen upper witness and policy.

certification. The fitted witness is independent of the policy and valid over the full original wealth and action ranges. Its loss of wealth-specific information is economically consequential. A separate wealth-sensitive linear-programming pilot is retained with its computation log and rejected candidate diagnostics; an unvalidated collocation objective is not used in either table.

APPENDIX G: VERIFICATION AND COMPUTATIONAL ERROR

G.1. Selection, stopped comparison, and the neural bridge

PROOF OF PROPOSITION 1: For a fixed jet the three controls enter additively. The consumption objective has derivative $c^{-u} - v_x$ and second derivative $-uc^{-u-1} < 0$. If $v_x \leq 0$ it is increasing; otherwise its unconstrained maximizer is $v_x^{-1/u}$. The adjustment objective is $\theta v_u - k\theta^2/2$, which is strictly concave for $k > 0$. The portfolio objective is $h_1 p + h_2 p^2/2$. A negative h_2 gives the projected vertex. A nonnegative h_2 gives a maximum at an endpoint, including the linear and constant cases. Compactness and the stated tie rule give a feasible measurable selection. These are control-space statements at a fixed jet, not parameter-space convergence statements. *Q.E.D.*

PROOF OF THEOREM 1: Localize the process on interior compact sets and apply discounted Itô's formula to an upper witness on each time slab. The discounted terminal witness plus accumulated reward equals the initial witness, the expected integral of $R_a U$, and the expected sum of discounted forward jumps. The latter sum is nonpositive. Boundary comparison gives $g \leq U + b_U$. Nonnegativity of discounting and the deterministic bound $\tau \leq T$ therefore imply

$$J_k^a(t, s) \leq U(t, s) + b_U + \epsilon_U \mathbb{E} \int_t^\tau e^{-\rho(r-t)} dr \leq U(t, s) + b_U + C_\rho(T-t)\epsilon_U.$$

Uniform integrability permits removal of localization. Taking the supremum over all admissible controls proves the optimal-value bound; attainment is unnecessary. Apply the same identity to L

along the implemented policy. Its forward jumps are nonnegative, $g \geq L - b_L$, and $R_\pi L \geq -\epsilon_L$, giving the lower bound. Subtraction proves the regret statement. The process is stopped at the economic exit; there is no use of an unproved interior trace of the optimal value. *Q.E.D.*

PROOF OF THEOREM 2: The cover and representative intervals imply $|R_\pi v| \leq e$ and $0 \leq \Gamma \leq q$ everywhere. Hence $\sup_a R_a v = R_\pi v + \Gamma \leq e + q$ and $R_\pi v \geq -e$. Apply Theorem 1 with $U = L = v$, $b_U = b_L = b$, $\epsilon_U = e + q$, and $\epsilon_L = e$. A derivative bound with respect to the chosen cell norm gives the two moduli by integration along segments, or along a specified path inside the cell. In a nonconvex cell, a verified path-length radius must replace the Euclidean radius. The vanishing-allowance assertion follows directly from the bound and does not require a compact set of network parameters. *Q.E.D.*

The finite counterpart follows from positivity of the continuation operators: propagate the evaluation error with one factor δ , propagate the optimality error with one factor δ , and add the candidate gain. Backward induction from terminal error b gives (8). The full state-dependent induction, correction rule, and recursive-operator versions are reproduced in the preservation supplement. Unlike this finite identity, a continuous interpretation additionally needs either verified operator discrepancy terms or an independent continuous verification route. The present original-payoff calculation uses the latter.

G.2. Arithmetic convention

The mathematical constants specifying the economy are exact decimals. Stored network weights, policy constants, and fitted coefficients are interpreted as exact binary64 numbers. Constant formation and polynomial coefficient conversion use 70-digit interval arithmetic; scalar optimization bounds use at least 50 digits. Vectorized interval addition, subtraction, multiplication, division by a strictly positive interval, and pairwise summation round both endpoints outwards with the adjacent binary64 number. An interval square may be enclosed by ordinary interval multiplication; this is conservative even when its lower endpoint is negative. No transcendental library approximation is treated as a rigorous interval merely because its output was subsequently rounded once. Exponentials in vectorized routines are replaced by Taylor polynomials with analytic remainders; interval logarithms, trigonometric constants, and square roots are formed in the multiprecision layer.

All derivative and residual claims concern the mathematical function represented by the stored constants, not the real-time floating-point implementation of an arbitrary neural policy. The original time controls are themselves stored constants, so their continuous implementation is unambiguous. External-network rollouts are explicitly floating-point performance estimates. This distinction is retained in every numerical table.

APPENDIX H: ORIGINAL-ECONOMY FITTED WITNESSES

H.1. Global fallback and localized upper bound

For $G = -.02(u - 2)^2 + .1 \log x$, consumption and portfolio selection give $c = .8$ and $p = .8$. The adjustment gain is bounded above by $.000512/k \leq .001024$ for $k \geq .5$. The preference part of $-\rho G$ is at most $.000512$. Consumption utility is maximized at the upper endpoint $u = 2.8$ on the stated domain. The portfolio drift and diffusion contributions at $p = .8$ are $.00352$. Maximizing $-.08/x - .004 \log x$ over $[.5, 2]$ gives $-.04 - .004 \log 2$. Consequently

$$\sup_a R_a G \leq \frac{.8^{-1.8}}{-1.8} + .000512 + .001024 + .00195 + .00352 - .04 - .004 \log 2 < -.8659.$$

Also $G \geq g$. Stopped verification gives $V_k \leq G$ globally on the original domain, including as a continuation bound at an artificial verification face.

PROOF OF PROPOSITION 6: The derivatives of $G + F$ in x coincide with those of G . Thus the same c and p selection applies. The residual after maximizing over wealth is precisely (65). For $z = 8h \exp\{\lambda(u - u_0) + \kappa h\}$, where $h = 1 - t$,

$$(\partial_t + \mathcal{L}^a - \rho)z = 8e^{\lambda(u-u_0)+\kappa h} \{-1 + h(-\kappa + \lambda\theta + .00125\lambda^2 - \rho)\}.$$

The choices $(|\lambda|, \kappa) = (120, 42), (160, 64)$ satisfy $\kappa = .2|\lambda| + .00125\lambda^2$. Both terms therefore have nonpositive killed generators for every admissible control. Moreover $(\partial_t - \rho)C_\rho(1 - t) = -1$. It follows that U has nonpositive residual inside the verification domain.

At $t = 1$, $F = Z = C_\rho = 0$, and $U = G$. At an actual wealth exit, $F \geq -8h$ implies $U \geq G - 8h = g$. At either artificial preference face, one summand of Z is at least $8h$, so $U \geq G$. Stop first at the economic exit or an artificial face. If the artificial face occurs first, the remaining original-model payoff is bounded by G under every admissible continuation, by the global fallback just proved. Concatenated stopped verification gives the asserted bound without modifying the original dynamics or opportunity set. *Q.E.D.*

H.2. Polynomial source enclosure and refinement

Write $u - 1 = A + Bz$, where $A = 1.05$, $B = .55$, and $|z| \leq 1$. Let $D = (A^2 - B^2)^{1/2}$ and $q = B/(A + D) < 1$. The uniformly convergent expansion

$$\frac{1}{A + Bz} = \frac{1}{D} \left[1 + 2 \sum_{n=1}^{\infty} (-q)^n T_n(z) \right]$$

has truncation error at degree 16 bounded by $2q^{17}/\{D(1 - q)\}$. If $L = -\log(.8)$, approximate $e^{L(A+Bz)}$ by e^{LA} times the degree-eight Taylor polynomial of e^{LBz} . Its remainder is at most $e^{L(A+B)}(LB)^9/9!$. Multiplying the two enclosures, with the relevant sup-norm bounds, gives the stored uniform error below 2.117×10^{-9} for $-.8^{-(u-1)}/(u - 1)$. This error is added to the residual upper bound, not fitted away.

The H_k branches are $-.2q - .02k$ for $q \leq -.2k$, $q^2/(2k)$ for $|q| \leq .2k$, and $.2q - .02k$ for $q \geq .2k$. On each box, interval Bernstein coefficients enclose q and every potentially applicable residual polynomial. A box touching a switching threshold includes both adjoining branches. The comparison uses outward threshold intervals; binary64 representation of $.2k$ cannot exclude a true branch. A tensor polynomial lies between its minimum and maximum Bernstein coefficients. De Casteljau subdivision preserves this enclosure and is repeated on boxes whose upper coefficient exceeds the prescribed ceiling. The independent check of $F/h \geq -8$ uses the same convex-hull property.

For a fixed polynomial whose true residual lies strictly below a ceiling, Bernstein refinement eventually verifies that ceiling, up to the explicitly accounted source and arithmetic allowances. This is a certification statement about a fixed candidate with slack. It is not a theorem that residual fitting, a wealth-independent ansatz, or arbitrary Adam training produces approximants converging to the optimal value. The executed ceiling sequence and all visited-cell counts are retained. The separate wealth-sensitive linear-programming pilot has not received this validation and contributes no upper endpoint to the reported results.

APPENDIX I: INDEPENDENT EVALUATION OF THE LEARNED TIME POLICY

The projected policy has constant c_j, θ_j on each of sixteen slabs, $p = 0$, and $0 \leq \theta_j \leq .2$. Training chooses these constants, but the subsequent evaluator reads only the frozen policy. The exact wealth formula in the main text and the outward budget check show $x_1 > .5$. Since $c_j > .025$ and $x_0 = 1.25$, wealth is decreasing throughout the horizon and never reaches the upper boundary. Neither Brownian correlation nor a wealth discretization affects this conclusion when $p = 0$.

Without preference stopping, $u_t = 2 + \mu(t) + .05W_t^u$, with $\mu(t) = \int_0^t \theta_r dr \in [0, .2]$. An upper-boundary exit requires $W_t^u \geq 12$ at some $t \leq 1$; a lower exit requires $W_t^u \leq -16$ and is also contained in $\{\sup_{t \leq 1} |W_t^u| \geq 12\}$. The exponential martingale maximal inequality on each side gives

$$p_{\text{exit}} \leq 2e^{-72} < 1.077 \times 10^{-31}.$$

The original preference diffusion is still stopped at exact first exit. Its rare-exit correction is not set to zero.

Put $r = u - 2$ and $L_j = -\log c_j$. Before exit, $|r| \leq .8$ and utility is $-e^{L_j} e^{L_j r} / (1 + r)$. Replace the reciprocal by $\sum_{m=0}^{80} (-r)^m$ and the exponential by its degree-ten Taylor polynomial. The geometric remainder is at most $.8^{81} / .2$; the exponential remainder is at most $e^{.8L_j} (.8L_j)^{11} / 11!$. The product bound, including the reciprocal and exponential suprema, yields a uniform error for the degree-90 polynomial P_j . This bound holds on the whole original preference interval, not just at training quadrature nodes.

For a Gaussian random variable with mean μ and variance v , its raw moments satisfy $M_0 = 1$, $M_1 = \mu$, and

$$M_n = \mu M_{n-1} + (n-1)v M_{n-2}.$$

Hence $\mathbb{E}P_j(u_t - 2)$ is evaluated by this recurrence with $v = .0025t$, and the terminal quadratic preference settlement has an exact Gaussian expectation. On each time rectangle the intervals for μ , v , discounting, and the polynomial coefficients enclose the entire integrand. Outward sums of interval range times interval length therefore enclose its integral. These ranges are not quadrature-error estimates inferred from two discretizations. The three resolutions are independent enclosures of the same integral.

To compare this free-Gaussian polynomial calculation with the stopped economic payoff, let E be the event of preference exit. Off E the two paths agree through T . On E , the omitted polynomial tail is bounded using

$$\mathbb{E}[\mathbf{1}_E |P_j(u_t - 2)|] \leq \sqrt{p_{\text{exit}}} \|P_j(u_t - 2)\|_2, \quad \|P_j\|_2 \leq \sum_m |b_{jm}| \sqrt{M_{2m}}.$$

Even Gaussian moments are bounded uniformly by evaluating their nonnegative coefficient formulas at $\mu = .2$ and $v = .0025$. The same argument bounds the free terminal settlement, while actual exit settlement has magnitude below 8.1. Discounted adjustment cost omitted after exit is at most $.02k\mathbf{1}_E$. One valid total stopping allowance is thus

$$\sqrt{p_{\text{exit}}} \left\{ \max_j \|P_j\|_2 + \|G(u_1, x_1)\|_2 \right\} + p_{\text{exit}} (.02k + 8.1).$$

The uniform utility approximation allowance is added separately, multiplied by $C_\rho(1)$. At $k = 2$ the stopping allowance is below 5.388×10^{-16} ; the utility approximation allowance is below 1.151×10^{-7} before multiplying by $C_\rho(1)$. The larger remaining width comes from the explicitly enclosed time integration, not from ignoring a boundary event.

The free adjustment budget is exactly $\sum_j (\theta_j^2/2) \int_{j/16}^{(j+1)/16} e^{-.04t} dt$. Stopping can reduce this quantity by at most $.02p_{\text{exit}}$. Its interval, and the strictly positive wealth-feasibility margin, are retained at full precision. Tight budget intervals describe these deployed policies, not optimal-budget identification.

APPENDIX J: ECONOMIC COMPARISON AND THE RESTRICTED DUAL

J.1. Cost ordering and value enclosures

PROOF OF THEOREM 4: For each policy, $J_k(\pi) = A(\pi) - kB(\pi)$ is affine and nonincreasing because $B(\pi) \geq 0$. The supremum is convex and nonincreasing. The two approximate optimality inequalities give

$$A_1 - k_1 B_1 \geq A_2 - k_1 B_2 - \varepsilon_1, \quad A_2 - k_2 B_2 \geq A_1 - k_2 B_1 - \varepsilon_2.$$

Adding and dividing by $k_2 - k_1 > 0$ gives (26). Zero allowances give budget ordering for any optimizers at distinct prices. If π_k is optimal at k , feasibility at k_+ implies $V(k_+) \geq V(k) - (k_+ - k)B(\pi_k)$; feasibility at k_- implies $V(k_-) \geq V(k) + (k - k_-)B(\pi_k)$. Substitute the appropriate value-enclosure endpoints and intersect with the primitive budget range to obtain (27). *Q.E.D.*

The access lower bound follows from $V_k \geq J_k$ and $V_R \leq \bar{V}_R$. Its nonnegativity follows because the restricted policy set is contained in the flexible policy set. Likewise $V_k - V_R \leq \bar{V}_k - \underline{J}_R$. These inequalities compare two optimal values even though the feasible policies supplying the lower bounds need not be optimal.

J.2. Cancellation of the portfolio and stopped localization

PROOF OF THEOREM 5: For Y as specified, quadratic covariation gives

$$d(Y_t x_t) = (.04Y_t x_t - Y_t c_t)dt + \text{local martingale}.$$

The $.06pYx$ portfolio drift cancels the $-.06pYx$ covariation term for every adaptive p . Including $-.5Y$, the discount term, and the preference quadratic yields

$$R_{c,p}W_0 = \frac{c^{1-u}}{1-u} - yc + .01y - .00005 - .004\log(.5) + .0008(u-2)^2 \leq S(u, y).$$

There is no W_0 mixed u - x or u - y derivative. Correlation is retained in the transition law of (U, Y) used to evaluate the source potential.

For a source lower bound use the feasible consumption .35: its utility on $u \in [1.5, 2.6]$ is bounded below by $-2/\sqrt{.35}$, while $-.35y \geq -4.2$. The remaining terms sum to a nonnegative quantity on this rectangle. Thus $S > -7.581 > -8$. For an upper bound discard $-yc$, maximize utility at $c = .8, u = 2.6$, use $.01y \leq .12$, and bound the remaining terms by .0033. This gives $S < -.7698 < 0$. The clipped extension is bounded in $[-8, 0]$, so $-8h \leq F_R \leq 0$.

The discounted Feynman-Kac potential satisfies the conditional expectation identity for its remaining source integral. Consequently $e^{-\rho r} F_R(h - r, U_r, Y_r) + \int_0^r e^{-\rho s} \tilde{S}(U_s, Y_s) ds$ is a martingale. Stopping this identity on a bounded verification rectangle is legitimate because the source is bounded. Inside the rectangle $\tilde{S} = S$, so adding the identity to discounted Itô verification for W_0 cancels its residual majorant. This argument does not require an assumed globally smooth optimal value, or differentiability of the clipped source at its outer faces.

Preference localization with $\theta = 0$ uses $\kappa = .00125\lambda^2$, giving 50 and 72 for $|\lambda| = 200$ and 240. For $l = \log Y$, $dl = -.025dt - .3dZ^x$. Each exponential localization term with $|\lambda| = 22$ has nonpositive killed generator because $22.33 = .025(22) + .045(22)^2$ dominates either sign of the log drift. As before, the factor $8h$ supplies at least $8h$ at its associated artificial face.

For $x \in [.5, 2]$ and $y \geq .2$,

$$W_0 - G = y(x - .5) + .1 \log(.5/x) \geq 0 :$$

the expression is zero at $.5$ and has derivative $y - .1/x \geq 0$. At an actual wealth exit the source term is at least $-8h$, so $W_0 + F_R + Z_R \geq g$. At a terminal time it is at least G . At any artificial u or y face a localization term pays for the entire possible source deficit, giving continuation value at least G . Actual preference exit cannot precede these inner preference faces. Invoke the global fallback $V_R \leq G$ after an artificial exit and apply the stopped identity before it. This proves (34) over all original adaptive consumption and portfolio controls. The value of Y_0 is an auxiliary witness parameter, not an imposed wealth price or policy restriction. *Q.E.D.*

J.3. Validated evaluation of the restricted witness

Let z_1, z_2 be independent standard Gaussians and put $r = v^2$. At the central state,

$$u = 2 + .05v\{-.25z_1 + \sqrt{.9375}z_2\}, \quad l = \log(1.55) - .025v^2 - .3vz_1.$$

The source expectation is integrated with weight $2ve^{-.04v^2}\varphi(z_1)\varphi(z_2)$ on $v \in [0, 1]$. The box $z_1, z_2 \in [-6, 6]$ lies strictly inside the stated u, y verification rectangle under this transformation. The omitted probability is at most $4e^{-18}$. Since the extended source lies in $[-8, 0]$, omission contributes an interval $[-8C_\rho(1)4e^{-18}, 0]$, whose width is less than 4.778×10^{-7} . The initial localization addition is less than 2.200×10^{-9} .

For completeness, the Hessian used by the evaluator is explicit. Let $c = \text{proj}_{[.05, .8]} e^{-l/u}$, $L = -\log c$, $y = e^l$, and let χ be one for an interior consumption solution and zero for a strictly constrained solution. Across a switching box use $\chi \in [0, 1]$. With $Q = c^{1-u}/(1-u)$,

$$\begin{aligned} Q_u &= \frac{e^{L(u-1)}\{1 - L(u-1)\}}{(u-1)^2}, & Q_{uu} &= -\frac{e^{L(u-1)}\{[L(u-1) - 1]^2 + 1\}}{(u-1)^3}, \\ S_u &= Q_u + .0016(u-2), & S_l &= -yc + .01y, \\ S_{uu} &= Q_{uu} + \chi yc L^2/u + .0016, & S_{ul} &= -\chi yc L/u, \\ S_{ll} &= -yc + \chi yc/u + .01y. \end{aligned}$$

The envelope is continuously differentiable across clipping surfaces; its gradient is locally Lipschitz and these piecewise second derivatives give a valid integral Taylor bound across such a surface. Differentiate the displayed (u, l) transformation by the chain rule to obtain a Hessian in (v, z_1, z_2) . Every intermediate expression is enclosed on the entire box. Bounds on the absolute Hessian entries give the usual one-half diagonal and full mixed-term Taylor remainder.

The constant and linear Taylor terms are integrated exactly for their polynomial approximation to the weight. Gaussian weighted moments through order two use 111 terms of $e^{-z^2/2}$ and remainder bounded on $|z| \leq 6$ by $e^{18}18^{111}/111!$. Time moments use thirteen terms of $e^{-.04v^2}$ with their analytic remainder. Source exponentials are evaluated with degree-40 Taylor polynomials on arguments in $[-3, 3]$, with error at most $e^3 3^{41}/41!$. All antiderivatives, remainders,

interval additions, and sums are rounded outward. The two diagonal second moments are non-negative; intersecting their enclosures with the nonnegative half-line is valid. Mixed moments are conservatively bounded by cell-radius products times probability mass.

The successive box counts are 16,384, 131,072, and 1,048,576. Their intervals enclose the same initial *dual witness*. Only an upper endpoint is used to bound the restricted optimum. Combining the final upper endpoint with the independently certified restricted-policy lower gives a true optimal-value interval of width less than .009047. The distinct width below .000320 describes integration of the witness, not the primal–dual gap.

APPENDIX K: CONTINUOUS ACTION AND ABSOLUTE-VALUE CERTIFICATION

K.1. The scalar global action bound and interval neural jets

PROOF OF THEOREM 7: For any action with $\sum_i a_i = z$ and any real λ ,

$$\sum_i f_i(a_i) = \sum_i \{f_i(a_i) - \lambda a_i\} + \lambda z \geq D(\lambda) + \lambda z.$$

The second derivative of $.3 \sin z$ is bounded below by $-.3$. For $z = z_0 + e$, $|e| \leq r$, its quadratic Taylor minorant gives

$$\lambda z + .3 \sin z \geq \lambda z_0 + .3 \sin z_0 + (\lambda + .3 \cos z_0)e - .15e^2,$$

which implies (48). Each scalar f_i has second derivative $(2 + .2 \cos 2s)/d \geq 1.8/d = m$. Thus $q = f_i - \lambda s$ is m -strongly convex. Its tangent quadratic at any feasible s_0 gives a lower bound $q(s_0) - q'(s_0)^2/(2m)$. At an endpoint, only a feasible descent direction is relevant: an outward-pointing minimization direction contributes zero, while an inward-pointing direction contributes the derivative magnitude. This follows either by minimizing the tangent quadratic over the half-line or directly from the sign of the linear term. The stated lower bound remains valid if the quadratic minimizer lies beyond the opposite endpoint, because relaxing the displacement can only lower it.

The sum of these scalar lower bounds is no larger than $D(\lambda)$. A feasible action gives a rigorous upper bound by direct interval evaluation. The scalar interval $[-d, d]$ covers every possible action sum. Subdivision, pruning only a box whose lower bound exceeds a feasible upper bound, and taking the minimum surviving lower bound preserve a global enclosure. Root finding affects efficiency but not validity. Finally $|H_p(a) - H_{\hat{p}}(a)| \leq 2 \sum_i \delta_i$ on the action box. The true candidate upper and true minimum lower can each worsen by at most this amount, giving the $4 \sum_i \delta_i$ gap correction. *Q.E.D.*

For an affine layer with stored weights, interval automatic differentiation propagates both activations and spatial derivatives. A tanh layer is evaluated as $(e^{2z} - 1)/(e^{2z} + 1)$ with a strictly positive denominator and derivative $1 - \tanh^2 z$, all in multiprecision intervals. The terminal embedding is differentiated as its mathematical formula, including the exact $\sqrt{d}$, not as a previously rounded autograd constant. This gives a componentwise interval for the derivative of the represented network. Subtracting the stored floating-point gradient gives the δ_i . The two executed tests use the inherited eight- and sixteen-dimensional learned checkpoints and the same jets as the action certificates. The summed errors are below 4.584×10^{-17} and 7.877×10^{-17} respectively. These are local jet checks, not whole-state residual bounds.

K.2. A lower bound retaining the nonquadratic terminal payoff

PROOF OF PROPOSITION 4: For $|a| \leq 1$, $\sin(a)/a \geq \sin 1$, with the continuous value at zero. Consequently $a^2 + .1 \sin^2 a \geq r_0 a^2$, where $r_0 = 1 + .1 \sin^2 1$. Also $.3 \sin(\sum_i a_i) \geq -.3$. Replacing running cost by $r_0 \|a\|^2/d - .3$ decreases every original control's cost; allowing all $a \in \mathbb{R}^d$ decreases the infimum further. The original terminal payoff q_d is unchanged.

The relaxed HJB equation is

$$v_t + \frac{\sigma^2}{2} \Delta v - \frac{d}{r_0} |\nabla v|^2 - .3 = 0, \quad v(1, x) = q_d(x), \quad \sigma^2 = .32.$$

Put $\lambda = \sigma^2 r_0 / (2d)$ and use the logarithmic heat representation. At $X_0 = 0$ it gives

$$v(0, 0) = -.3 - \lambda \log \mathbb{E} \exp\{-q_d(\sigma Z)/\lambda\}, \quad Z \sim N(0, I_d).$$

The bounded cosine and coercive quadratic terminal terms give finite positive expectations. The Gaussian heat solution is smooth before the terminal date and has at most quadratic growth. Standard localization of its verification identity, or truncation followed by monotone Gaussian integration, validates the relaxed lower bound without assuming a smooth solution of the original bounded-control problem.

Complete the square for the quadratic terminal component. Its Gaussian normalization contributes

$$\frac{.25r_0}{r_0 + 4} + \frac{\sigma^2 r_0}{4} \log(1 + 4/r_0) - .3.$$

Under the tilted Gaussian law the scalar sum divided by $\sqrt{d}$ has mean $m = 2\sqrt{d}/(r_0 + 4)$ and variance $s^2 = \sigma^2 r_0 / (r_0 + 4)$. Orthogonal coordinates disappear, leaving

$$v(0, 0) = \frac{.25r_0}{r_0 + 4} + \frac{\sigma^2 r_0}{4} \log(1 + 4/r_0) - .3 - \lambda \log \mathbb{E} \exp\{-(.2/\lambda) \cos(m + sZ_1)\}.$$

This is the representation in the proposition. An upper enclosure for the positive scalar expectation gives a lower enclosure for v , because $-\lambda \log$ is decreasing. That lower endpoint is therefore a lower bound for the original value. Q.E.D.

For the scalar integral let $\eta = .2/\lambda$ and $R = \lceil \sqrt{2(\eta + 50)} \rceil$. The omitted Gaussian contribution is no larger than $2e^{\eta - R^2/2}$. Interval ranges of the positive integrand times cell lengths enclose the integral on $[-R, R]$. The successive 1,024, 4,096, and 16,384-cell calculations add this analytic tail allowance and then apply the decreasing logarithm with outward rounding. Their widths measure integration uncertainty in the relaxed value. Relaxing the action bound and the running sine can leave a large structural gap even when this quadrature width is small.

APPENDIX L: REPLICATION AND SCOPE LEDGER

The central economic certificate is independent of the inherited killed-Euler discretization. Its fitted upper has a verified continuous-generator residual, an explicit source-polynomial remainder, a stopped-trace inequality, and exponentially bounded artificial-face continuation. Its feasible lower has exact deterministic wealth, interval Gaussian polynomial moments, an explicit utility approximation remainder, outward time integration, and an exact-exit probability correction. The restricted upper cancels continuous portfolio controls algebraically and evaluates

a bounded-source stochastic potential with Taylor, Gaussian-tail, and localization allowances. These routes do not establish the still-uncomputed operator discrepancy budget for the old finite scheme.

The external experiment has a different status. Each declared sample-and-hold policy is a feasible continuous control in the stated additive-noise dynamics. Real-arithmetic transitions are exact per holding slab, but the stored path estimates use floating-point network evaluations and a finite Monte Carlo sample. The rigorous lower bound supplies an absolute reference; subtracting it from a sampled mean does not create an outward policy-regret certificate. Paired path errors, seed variation, completed-update clock overshoot, setup, and warmup are recorded separately. The exact sign test concerns the frozen joint training-and-audit pipeline under independent seed draws. It does not certify an underlying PDE or imply that the six panels represent six independent problem families.

Every new table is generated from raw JSON or path records; display endpoints are rounded in the conservative direction. The frozen external protocol, all tuning trials, all 72 holdout pairs, all checkpoints, and all path arrays are retained. The validation program checks coverage of the declared panels, matching initialization hashes, pinned author revision, finite outputs, refinement ordering, primitive feasibility, the positive access lower bounds, and the explicitly false flexible .01 precision flags. The complete R8, R4, and delivered R5 textual and numerical material is included in the preservation supplement, and earlier source files remain byte-for-byte unchanged. Neither a failed pilot nor a historical manufactured-payoff test is used as new original-economy evidence.

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